Beyond Boundaries: Enhancing Exoplanet Discovery by Machine Learning

Tanisha Das, Dr. Chandra J
Department of Computer Science
CHRIST (Deemed to be University)

ABSTRACT

The quest for discovering habitable exoplanets has been a longstanding and highly debated topic in astronomy. Traditional exoplanet identification methods—such as the radial velocity or "wobble" method, direct imaging, and gravitational microlensing—demand extensive labor, time, and financial resources. Moreover, these techniques are often constrained by the limitations of current astronomical observatories. In recent years, machine learning and deep learning have revolutionized numerous scientific disciplines, including astronomy. Data scientists have been particularly inspired by the wealth of light curve data generated by space missions like the Kepler mission and the Transiting Exoplanet Survey Satellite (TESS). These missions provide critical data for examining light curves to detect planetary transits, which are the key indicators for identifying and confirming exoplanets—planets that reside outside our Solar System. This study offers a comprehensive review of the advancements and applications of both supervised and unsupervised learning methods in exoplanet research. It delves into the practical application of notable supervised learning algorithms, including Random Forests, Gradient-boosted trees such as LightGBM, and Artificial Neural Networks. The focus is placed on critical processes such as feature selection, dataset preprocessing, and model evaluation. Additionally, a comparative analysis of supervised machine learning algorithms is conducted using the Kepler Objects of Interest (KOI) cumulative catalogue. The experimental component focuses on classifying KOIs into planetary candidates and false positives using catalogue-derived features. Various machine learning models and a Multilayer Perceptron are evaluated using group-aware data splitting and cross-validation to assess their classification performance and generalisation. Overall, this paper highlights the potential of integrating machine learning techniques with astronomical data analysis to support the classification and assessment of exoplanet candidates.

I. INTRODUCTION

Planet detection has been a primary research subject in astrophysics since the finding of the first exoplanets. Exoplanets have been identified using a variety of approaches, including the radial velocity method, astrometry, direct imaging, and gravitational microlensing. The wobble method discovers exoplanets by detecting a Doppler shift in the star's light frequencies induced by its planets; direct imaging captures photos from exoplanets using extra-terrestrial observatories; and gravitational microlensing examines the distortion of the background light. These old procedures are time-consuming, expensive, and sensitive to measurement fluctuation. The transit method has been used to discover most verified exoplanets. A planet may migrate in front of its host star, blocking a portion of its light and causing its observed visual brightness to decline by a small percentage, depending upon the relative sizes of the host star and its planet. The host star's brightness is thus continuously examined for the transit method, and 'dips' in the resultant light curve are sought. The initial step in looking for transiting exoplanets is to see if the light curve has a train of transits. The first stage in analyzing recorded light curves is to look for periodic signals resembling transiting planets, known as "threshold crossing events" (TCEs). Human specialists perform this screening step either manually or through automated approaches. A lot of exoplanets have been discovered using the transit method, with the Kepler Space Telescope playing an important role. The Kepler pipeline, which processed data, identified transit signals, and validated their existence, was critical to the mission's success. Many search techniques exist to discover these periodic events in a time series of data, but determining which looks transit-like is typically done by eye. Exoplanet surveys such as Convection, Rotation, and Planetary Transits (CoRoT), Kepler, PLAnetary Transits and Oscillations of stars (PLATO), and Transiting Exoplanet Survey Satellite (TESS) discovered or will discover more signals in their data than a small team of individuals can quickly evaluate consistently. Machine learning techniques come in many varieties, some of which have already been applied to assess transit light curves. K-nearest neighbors, decision trees, random forests, and self-organizing maps (SOM) are a few examples. These strategies determine whether a signal appears to be a transit. This problem can be expressed in terms of dimensionality reduction and clustering. Thousands of points are used to characterize the contour of the feature in the light curves. However, only a few dimensions are required to indicate whether the signal exhibits a transit's steep ingress and egress and a flat profile outside the transit event. The key here is to classify the light curves in such a way that a dimensionality reduction process separates signals

that appear to be transits from those that do not appear to be transits. One disadvantage of this method is that transits can only be witnessed when the planet's orbit aligns precisely with the observer's location. Another concern is the high rate of false detections, such as through eclipsing binary systems or transits by planet-sized stars, necessitating alternate methods to corroborate discoveries. Sinusoidal fluctuations induced by spots, pulsations, tidal binaries, or contact binaries are the most prevalent type of false alert.

One key issue in evaluating exoplanetary data occurs when dealing with unknown transit duration. Only when the folding time matches the transit period can the folding approach, which averages flux data across particular periods, improve transit signals. Achieving high-resolution folding trial periods is critical for detecting novel transits with unknown periods. To address this issue, a two-dimensional convolutional neural network (2D-CNN), usually used for pattern recognition, has shown potential. NASA's Kepler mission and its follow-up survey initiative K2 constituted a big step in building a comprehensive inventory of exoplanetary systems. The database has the potential to provide vital insights into the principles underlying planet formation as well as the occurrence of Earth-like analogs suited for habitation. While Kepler was planned as a statistical endeavor to investigate the event of Earth-sized exoplanets inside habitable zones, early findings focused on individual systems, resulting in relatively homogeneous catalogs. When Kepler's data was introduced, many high-quality candidates emerged. However, these candidates mainly focused on faint host stars with V magnitudes more significant than 14. The faintness of these stars makes thorough follow-up using radial velocities impossible, especially for candidates with long orbital periods and low signal-to-noise ratios. In response, vetting procedures were expanded to include planet 'validation,' which involves statistical confirmation of a planet without additional evidence. While statistical validation is not as good as autonomous discovery methods, it has validated over 1200 planets, accounting for more than half of Kepler's discoveries. This validation strategy considered factors such as the multiplicity boost and an explicit probability estimate for each false positive scenario. These strategies have proven extremely useful not just for confirming planets but also for assigning follow-up resources efficiently. These validation approaches were still used even when follow-up was possible for brighter stars. The Planet Analysis and Small Transit Investigation Software (PASTIS), Blender, Virtual European Solar and Planetary Access (VESPA), the recently introduced Tool for Rating Interesting Candidate Exoplanets and Reliability Analysis of Transits Originating from Proximate Stars (TRICERATOPS), and a customized approach for Kepler's multiple planetary systems are among the planet validation procedures published in the literature. While each technique has advantages and disadvantages, VESPA stands out since it applies to many candidates. Using a single method, such as VESPA, to validate about 30% of known exoplanets raises risks across multiple interdependent exoplanetary research fields, including planet creation, evolution, population synthesis, and occurrence rates. Despite breakthroughs, the validation process for Kepler mission candidates is a continuing activity, requiring significant time and effort to remove false-positive identifications while revealing fresh candidate discoveries methodically. In April 2018, NASA launched the Transiting Exoplanet Survey Satellite (TESS) mission as a successor to the Kepler project, with the primary goal of surveying some of the cosmos' brightest stars. TESS creates around one million light curves per month while covering 85% of the sky, a coverage area 400 times bigger than Kepler. TESS is distinguished by its observation of stars that are substantially brighter (30 to 100 times) than those observed by Kepler, allowing the selection of targets suitable for comprehensive follow-up investigations using space- and ground-based telescopes. TESS's early discoveries include a Super-Earth in the Pi Mensae system, a planet with an ultra-short orbital period orbiting a red dwarf, and an Earth-sized exoplanet in the habitable zone. Neural networks have proven useful in planetary research, particularly multi-planet prediction and atmosphere categorization. Kipping and Lam, for example, used a neural network to forecast multi-planet systems. Although this neural network's primary goal was not to identify transits directly, it expertly analyzed the relationships between orbital period, planetary radius, and mass to predict the presence of extra celestial bodies within a planetary system. Given the constraint that TESS data primarily cover short-period planets (i.e., P 13.7 days), predictions for planets with longer periods can be effectively predicted by utilizing Kepler data correlations.This literature review examines machine learning techniques used in exoplanetary research, with particular emphasis on transit detection, candidate classification, and validation. It discusses the advancements, challenges, and opportunities associated with applying these techniques to astronomical datasets. In addition, this study presents a comparative evaluation of machine learning approaches using the Kepler Objects of Interest (KOI) cumulative catalogue to classify planetary candidates and false positives.

II. Literature Review

Christopher J. Fluke and Colin Jacobs explore the implementation of Artificial intelligence (AI) and machine learning (ML) in astronomy. The authors thoroughly examine the existing scene, concentrating on the degree of integration, breakthroughs, and revolutionary impact of ML and AI in astronomical research and allied disciplines. They examine the development and effectiveness of these technologies by systematically evaluating their capacity to meet difficult astronomical tasks ranging from data analysis to predictive modeling. The study sheds light on the many methodologies, algorithms, and models used at the junction of Machine Learning (ML), Artificial Intelligence (AI), and astronomy, revealing how these technologies are used to process massive astronomical datasets, find trends, optimize observations, and extract useful results. Data-centric methodologies integrating

statistical approaches, machine learning (ML), artificial intelligence (AI), and effective database systems are increasingly driving scientific discoveries in astronomy. This fusion speeds up the multidimensional process of discovering patterns inside datasets, which includes stages such as data selection, purification, processing, and reformation into actionable knowledge—a process similar to knowledge discovery in databases. The expected inundation of data from planned observatories and space missions such as the Large Synoptic Survey Telescope, Euclid satellite, MeerKAT, and Square Kilometre Array is driving the incorporation of ML and AI into the astronomical study. These cutting-edge technologies are crucial in classification, regression, forecasting, and discovery across astronomical data formats, including pictures, spectroscopy, photometry, time series, and catalogs [1].

The International Astronomical Union defines a planet as having three fundamental properties: orbiting the Sun, having enough mass to form a roughly spherical shape, and being the major gravitational object in its orbit. This description, however, does not fully apply to exoplanets, which are planets outside our Solar System. Exoplanets have less than 13 Jupiter masses but substantial enough not to be categorized as dwarf planets. The initial exoplanet discovery is still being debated, with multiple sources crediting it to different researchers and dates. Surveys for transiting exoplanets began with ground-based facilities such as Wide Angle Search for Planets (WASP) and Hungarian-made Automated Telescope (HAT). They were later supplemented by space-based missions such as the Kepler Space Telescope. The Kepler mission was essential in discovering possible exoplanets and compiling the Kepler Objects of Interest (KOI) catalog. The Hubble Space Telescope, the Convection, Rotation and Planetary Transits (CoRoT) satellite, and the Transiting Exoplanet Survey Satellite (TESS) are other telescopes active in exoplanetary research. TESS, launched in 2018, attempts to detect several planet candidates using transiting photometry. Astronomers have devised several ways to identify exoplanets and deduce various attributes, albeit indirectly, such as the composition of the exoplanet's atmosphere, temperature, and masses of both the exoplanet and its host star. Combining approaches can improve exoplanet and star characterization. The Transit Method, in which an exoplanet passes between the observer and its circling star, produces a transit event similar to a solar eclipse. Examining transit events entails analyzing light curves, which describe the light intensity from the observed star with time. When an exoplanet passes in front of a star, the brightness of the light curve decreases, indicating a possible transit occurrence. The finding of the planet HD 209458b in 1999 was the first successful application of this technology. Transit events have a small effect on stellar flux, usually around 1% for hot Jupiters and 0.01% for Earth-like planets. This approach can compute parameters such as transit depth, length, and planet-to-star size ratio. Under certain conditions, an analytic solution exists to determine stellar and planetary parameters from a planet transit light curve. This method, however, is limited to studying planets with orbits almost edge-on to the observer's line of sight or those very close to their stars. Furthermore, it favors detecting bigger exoplanets concerning their home stars. Noise in light curves presents obstacles and probable false positives in the exoplanet hunt. Multiresolution Analysis (MRA) is recommended as a significant method for exoplanet research, particularly during the detection and identification stages. MRA relies heavily on wavelets, described as functions that localize signals in location and scaling. In exoplanetary research, several wavelet models have been used. Masciadri and Raga in 2004, for example, used wavelet analysis to detect exoplanets in imaging studies. The Morlet wavelet was used by Bravo et al. in 2014 to analyze light curves and identify temporal variations associated with events such as transiting exoplanets. Wavelets provide an in-depth interpretation of light curves, exposing various physical events inside the signal. Multiple Machine Learning techniques were used in this study, and a performance analysis table was generated which can be found in the Result Analysis section of this paper [2].

Kyle A. Pearson et al. used simulated training data to teach deep neural networks how to anticipate single planetary transits in noisy photometric data. The generated data closely resembles that of a genuine planetary search survey. Deep neural networks are used to estimate the likelihood of prospective planetary signals in previously unobserved data after they have been trained. For training, 311,040 transit and non-transit samples were created. The training data are derived from a parameter space discretely sampled nine-dimensional hypergrid. The parameters limit transit duration to 30 minutes to 4 hours, which covers a significant percentage of the time domain. By using various systematic forms and transit sizes, the parameters are intended to mimic data from real-world search surveys. This method allows the deep neural network to distinguish between transit and non-transit events. Before being fed into the deep neural networks, the synthetic data is normalized to unit variance, and the mean is subtracted. The system is modeled using an artificial neural network, which mimics the biological way a brain solves issues. It is made up of interconnected neuronal units that have customizable connections and synapses dependent on activation states. The networks are made up of layers of neurons known as limited Boltzmann machines, each with input parameters and weights representing the relevance of the input parameters. Neurons in a network change input data in order to abstract information for the following layer. A neuron also has a bias, which acts as a threshold for the neuron's ultimate choice, whether it is a yes/no decision or a probability. For each deep network, a fully connected multi-layer perceptron (MLP) is used. A perceptron, a form of neuron, for example, accepts multiple inputs (e.g., photometric data) and produces a single output by weighing the inputs. The output is decided by whether the weighted total of the inputs exceeds a bias parameter-defined threshold value. The weighted sum $\sum_j w_j x_j$ determines whether the neuron's output is less than or larger than a certain threshold value, which is determined by the bias, b. The threshold, like

the weights, is a real value that is a neuron parameter. To state it in more mathematical terms:

$$\text{Output = 0 if } \sum_j (w_j x_j + b) \leq 0$$

$$\text{Output= } \sum_j (w_j x_j + b) \text{ if } \sum_j (w_j x_j + b) > 0$$

To identify the neuron output, activation functions such as the rectified linear unit (ReLU) are used to modify the input data with a weighted sum. By not overloading the activation and allowing efficient learning, the ReLU function addresses the 'exploding/vanishing gradient' problem that is typical in saturated activation functions such as sigmoid and tanh. Stochastic gradient descent (SGD) and Nesterov momentum are used to optimise the weights. Dropout, which randomly zeros a neuron's output, is used during training to prevent overfitting. A grid search was used to find the architecture factors and configurations that effect performance, and the best parameters were chosen. The best deep neural network employs 13,937 trainable weights spread across 105 neurons with 2,492 neural connections or synapses [3].

Dalya Baron presents a practical description and overview of the use of machine learning in astronomy. The author examines machine learning basics, its use in astronomy, common algorithms, data pretreatment procedures, and evaluation methodologies. The relevance of knowing supervised and unsupervised learning, as well as their applications in classification, regression, clustering, and anomaly detection tasks, is emphasized in the study. Furthermore, it emphasizes the importance of adequate dataset preparation, feature selection, scaling, and dealing with imbalanced data when it comes to generating relevant and reliable results in astronomy using machine learning. Using specified examples, supervised machine learning algorithms are used to determine a link between a set of measurements and a target variable. Once established, this relationship allows for predicting the target variable given previously unobserved data. The difference between traditional model fitting methods and supervised learning algorithms is that the former uses a predetermined model, whereas the latter builds the model based on the input dataset. Supervised learning algorithms are intrinsically capable of describing complex nonlinear relationships between observations and the target variable, potentially outperforming traditional model-fitting methods that rely on predefined models. The dataset in the domain of machine learning consists of objects, each of which has measured attributes and a target variable. Objects in astronomy are typically physical entities like as stars or galaxies, with characteristics indicating qualities such as spectra or light curves, or higher-level derived quantities from observations such as variable period or stellar mass. The type of target variable used depends on the task at hand: discrete for classification tasks, such as classifying spectra into stars or quasars, and continuous for regression tasks, such as predicting redshift using photometric observations. Model parameters are frequently estimated from data in supervised learning techniques. These parameters are an essential component of the learning model and must be used when making predictions. Furthermore, supervised learning techniques frequently use model hyper-parameters that are set outside of the model using various heuristics. The use of supervised learning algorithms consists of three stages: training, validation, and testing. The training stage entails configuring model hyperparameters and learning the model and its parameters from a subset of the input dataset known as the training set. The validation set is used to optimize hyper-parameters during the validation step. Finally, in the test stage, the trained model is evaluated on a previously unseen subset—the test set. Evaluation metrics are important in optimizing supervised learning algorithms since they are used for hyper-parameter tuning, assessing final model performance, feature selection, and algorithm comparison. Mean Absolute Error (MAE) and Mean Squared Error (MSE) are two frequent evaluation metrics in regression projects. Metrics like as Classification Accuracy, Confusion Matrix, and Area Under the Receiver Operating Characteristic Curve (ROC) Curve are often used in classification tasks. The ROC curve plots the True Positive Rate versus False Positive Rate to visualize algorithm performance in binary classification tasks. Unsupervised learning uses statistical tools such as clustering, dimensionality reduction, visualization, and outlier identification to extract knowledge from datasets. These technologies are essential in astronomy, particularly with the abundance of complicated, high-dimensional data from recent astronomical surveys. The report emphasizes the importance of these tools in producing discoveries and comprehending data patterns. The author describes how to estimate distances between objects in a dataset using Random Forest in an unsupervised context. They construct a useful distance metric, particularly effective for astronomical spectra, catching information from emission and absorption lines, by creating a synthetic dataset and training the Random Forest to detect covariance. The method is demonstrated for tracing the physical properties of celestial objects. The application of machine learning methods in Astronomy has become prominent in recent years. Basic supervised learning topics were covered, emphasizing input dataset selection and preprocessing, evaluation metrics for supervised algorithms, and a brief description of three popular algorithms: Support Vector Machine (SVM), Decision Trees, Random Forest, and shallow Artificial Neural Networks. The emphasis was on unsupervised learning approaches, which were roughly classified as clustering analysis, dimensionality reduction, and outlier detection. The supervised scenario, in which machines are trained for classification or regression based on existing scientific knowledge, is the principal application of machine learning in Astronomy. Unsupervised learning methods, while less frequent in astronomy, offer the ability to mine datasets for fresh information and promote new discoveries [4].

Susan Thompson et al. propose a unique machine-learning technique to identify transit-shaped signals in astronomical

data. The Kepler mission gathered time series data for about 160,000 stars, totaling 17 quarters of data, with each quarter lasting approximately 90 days. The observations were made at a rate of approximately 29.4 minutes per minute. The main goal is to find Threshold Crossing Events (TCEs), periodic signals consistent with transit and a major indicator of probable exoplanets. The Kepler pipeline's Transit Planet Search (TPS) component searches for these transit signals by whitening the data and seeking significant detections at various dates and transit durations (The Kepler mission was first launched in March 2009, and its primary mission, dubbed "K1," continued until May 2013. Following that, Kepler resumed its mission in an extending phase known as "K2," which began in 2014. Throughout the K2 phase, the spacecraft observed a different area of the sky every three months, known as a "campaign." Each campaign generated a new quarter's worth of time series data. Kepler had completed several campaigns, yielding over 27 quarters of data. During its mission, Kepler measured the brightness of nearly 200,000 stars). Once a probable transit signal has been detected, it is validated within the Kepler pipeline. It is specifically treated to the Data Validation (DV) module, where it is fitted with a transit model. The points reflecting the transit period are deleted throughout this fitting operation. Using two unique detrending methods, the pipeline evaluates each Threshold Crossing Event (TCE). The most important technique involves the use of a DV-median detrender. It starts by fitting and removing a harmonic series of sine waves, then applies a median detrender with a timescale set by the signal's duration. This procedure is designed to find transit-like features in the data being analyzed. When constructing the filter, the alternative detrender, the penalized least-squared (LS) approach, only considers the points outside the transit period. This approach improves the transit-like appearance of most signals by omitting in-transit points during detrending. Following detrending, the light curve's median value is normalized to zero, and deviations are reported as fractional changes in observed brightness. To begin analyzing the form of each probable transit in the data, the time series is folded into N points, and the event is placed at a phase of 0.5. These N points serve as the initial number of dimensions for each TCE. The goal is to describe the shape of the light curve evenly across all TCEs, requiring the same number of data points for each. Following that, the folded light curve is divided into bins. Most transits appear as a rapid decrease in the star's brightness, occurring in a relatively short period compared to the complete orbit. Since the form of the discovered brightness dip is more important information than its duration, bin bounds are set to guarantee that all transits cover nearly the same area. The timescale of the fascinating phenomenon determines the appropriate bin count. Approximately one-third of the binned sites are chosen to cover transit durations of five transits on either side of the reported event. The remaining two-thirds of the binned points, excluding those at 0, 0.5, and 1, are uniformly distributed throughout phases. This intentional removal of binning points at phase 0 reduces the influence of important secondary signals on specific TCEs while having little effect on evaluating the out-of-transit light curve form. As a result, TCEs with significant secondary signals retain a transit-like look. The binned data are sorted by phase and normalized so that the smallest measured binned value within in-transit bins is set to -1. This binning method keeps the core form of the transit while removing details such as the depth and duration of the dip. After this, the goal is to simplify the decision-making process of whether a binned light curve mirrored a transit. As a result, the possibility of dimensionality reduction was investigated. While many procedures attempt to keep several dimensions in a dataset, the technique chosen was Locality Preserving Projections (LPP). Because of its unambiguous mapping across higher-dimensional space, LPP outperforms nonlinear methods such as Locally Linear Embedding (LLE). Despite being trained entirely on the original dataset, this quality enables precise evaluation even on inserted signals. LPP resembles Principal Component Analysis (PCA), a popular linear dimensionality reduction method. However, during dimensionality reduction, LPP strives to preserve local neighborhoods as measured by Euclidean k-nearest-neighbors. As a result, LPP is less sensitive to outliers than PCA and can map extremely nonlinear manifolds, a feat also accomplished by nonlinear approaches. The LPP algorithm takes the N binned points for a TCE sample and maps them to fewer dimensions, n while attempting to keep neighboring elements as specified by k-nearest neighbors. Then, the distance to an integer number, k, of the labeled data set's nearest known transits is computed. MATLAB's knn-search method is used to determine a Euclidean distance using a Minkowski distance with an exponent of two. The LPP transit metric is the mean of these k distances. Transit-like TCEs are as close to recognized transits as they are to each other. The metrics' performance depends on a well-labeled dataset of known transit-like events for training. A collection of previously validated Kepler Project Threshold Crossing Events (TCEs) helped to methodically classify 5678 TCEs as transit-like, out of which 3738 are planetary candidates, 633 have a large secondary, 1307 are confirmed false positives, and 1039 are not transit-like. This dataset was used to develop the Linear Programming Problem (LPP) transit metric, which is used to determine whether a TCE is a planetary candidate [5].

Chintarungruangchai and Ing-Guey Jiang investigate the ability of sophisticated deep-learning models to detect transits in noisy light curves. Traditional approaches frequently fail to confirm the presence of transits immediately, necessitating later validation and vetting steps. Deep learning models can be adjusted for transit detection or vetting based on the input data and training methodologies. By training and testing the model on simulated light curves with modeled red noise, Zucker and Giryes demonstrated the usefulness of 1D-CNN for identifying exoplanet transits. 1D-CNN was advocated as a viable technique for quickly evaluating large amounts of data from space missions in transit investigations. Giryes advocated using 1D-CNN for planetary transit identification

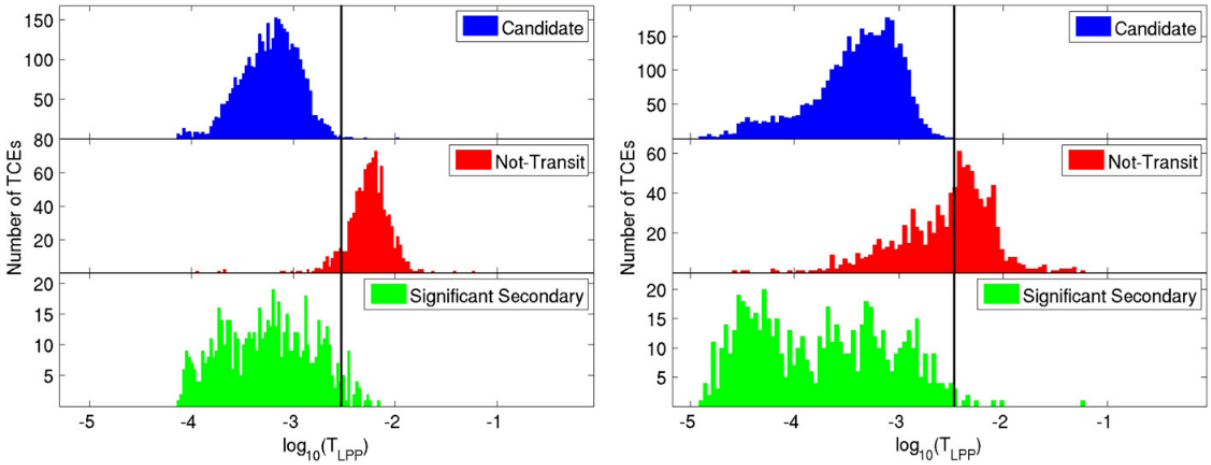


Fig. 1 Histogram of the labeled data set's LPP transit measure using the DV median (left) and penalized Least-squared (right) detrenders. To help direct the eye, a three-sigma line referenced in the text is depicted in black. To demonstrate the somewhat higher failure rate among binaries, the transit-like population has been divided into candidates and significant secondaries [5].

2018, stressing its potential with hypothetical light curves, including modeled red noise. These synthetic datasets were used to train and test the algorithm, resulting in the recommendation of 1D-CNN as a potential technique for analyzing large amounts of space mission data in future transit investigations. Pearson and his colleagues used the phase folding approach to capitalize on transit periodicity in 2018. The transit signal is enhanced after folding, improving transit detection efficiency. However, this technique presents a barrier when searching for transits in future observational light curves because prior knowledge of the transit period is required. Therefore, the 2D-CNN models were created in response to this challenge. Five deep-learning models are built to understand the benefits of 2D-CNN models better. Model 1 is a multilayer perceptron (MLP) with three fully connected layers. Models 2-5 are CNN models with four convolutional layers. In 2D-CNN models, the kernel size and stride are represented as k1xk2. Model 2, dubbed "1D-CNN-folding-0," is a 1D-CNN model that uses the whole light curves without folding. Similar models have been utilized in previous works. The basic structures of the convolutional layers follow a specific approach, differing mainly in the number of input data points and containing three completely connected layers. Model 3, dubbed "1D-CNN-folding-1," is a 1D-CNN model that employs folding. The mean of the folded light curves is computed and utilized as input data. Model 4, dubbed "2D-CNN-folding-1," is a 2D-CNN model incorporating folding. The second dimension's kernel size and stride are the same as in Model 3. Model 5, also known as "2D-CNN-folding-2," is a variant of Model 4. Except for using two data periods for each folding, all specifications remain the same as in Model 4. Because few stars are spotted with short cadence, only long cadence data Q1-Q17 from the Kepler data are included in this investigation. Multiple columns of light-curve data are present, including two types of measured fluxes. Simple aperture photometry (SAP) is one kind, which represents the flux obtained through direct photometric analysis but may still contain artifacts. The other is pre-search data conditioning (PDC), which removes artifacts from the data (Smith et al., 2012). PDC data is used. Outliers caused by astronomical phenomena, such as star flares and microlensing events, are not removed from PDC light curves. For effective model training, half of the data must be transit light curves, with the other half being non-transit light curves. The flux decrease characteristic of transit signals is added to Kepler non-transit light curves to construct synthetic transit light curves that approximate identified planets with a comparable size distribution and size range. Kepler non-transit data is used to generate the non-transit light curves. This method ensures that the deep-learning models are trained on equal transit and non-transit light curves. Several parameters are involved in this process, including the planet radius to stellar radius ratio $\frac{r_p}{r_s}$, the orbital semi-major axis to stellar radius ratio $\frac{a}{r_s}$, inclination i, transit period τ, linear limb darkening coefficient u_1, and quadratic limb darkening coefficient u_2.

$$\text{SNR} = \frac{(r_p r_s)^2}{\sigma_{lc}} \cdot \sqrt{n}$$

where σ_{lc} is the standard deviation of the flux values of a normalized light curve, and n is the number of observed points during the transit duration. By entering values for these parameters, transit light curves are generated. The SNR is set to be between 7 and 56 and is distributed uniformly in log(SNR) space, deciding r_p/r_s. The orbital period (transit period) τ is set to be uniformly distributed in log(τ) space and is used to calculate the value a/r_s. The coefficients u_1 and u_2 have normal distributions, with the mean and standard deviation assumed to be the same as in Kepler transit data for u_1 and u_2. The inclination i is evenly divided between 85 and 90 degrees. Using these parameters, the transit duration is determined, omitting factors that result in

a transit period shorter than 0.04 day, which is two times the observing cadence of Kepler long cadence data. The data was divided into three sets: training, validation, and testing. During training, model parameters were continuously updated using Stochastic Gradient Descent (SGD), with mini-batches of 64 light curves given successively. The technique was repeated over several epochs while accuracy was monitored. After a defined number of epochs, the models were settled, and their accuracy was verified using testing data. Regarding data distribution, 80% was training data, 10% was validation data, and 10% was testing data. The proportion of transit and non-transit samples was balanced, and accuracy improved as the number of light curves grew. Then, it also looked into how different folding periods and signal-to-noise ratios (SNR) affected model performance. CNN models with folding performed well across a range of phase range values and SNR levels. The effect of the folding period different from the transit period was also investigated. The accuracy of the models varied, underscoring the need to match these times for precise forecasts. Overall, the study showed that deep learning models successfully detect transits in light curves, with CNN models with folding offering good accuracy and robustness to diverse parameters [6].

Abhishek Malik et al. discovered an innovative technique that was taken to build a light and efficient automatic vetting system, to make potential breakthroughs in the domain of searching for exoplanets. A binary classifier is being built to divide each time-series photometry, known as a light curve, into the separate categories of ”planet candidate” and ”non-candidate.” A key distinction between deep learning approaches is that the latter has the potential to extract features autonomously, whereas classical machine learning requires prior feature computation, which is supplied as model input. Time Series Feature extraction based on scalable hypothesis tests (TSFresh), a Python-based library, is used for feature extraction. The K2 photometry was obtained from the Mikulski Archive for Space Telescopes (MAST) and calibrated using Vanderburg and Johnson's approach. Iteration is used to eliminate typical stellar-type-dependent variability, which includes data smoothing, binning, cubic spline fitting to the binned data, and the removal of negative 3σ outliers. Transit signals are injected at random into half of the analyzed light curves. Following a blind search for eclipsing binary systems facilitated by adopting the Box-fitting Least Squares (BLS) approach, known planet systems are rigorously removed from the data sample. To maintain a uniformly sampled time series, gaps in K2 light curves are filled by interpolation. Any light curve with a measured signal-to-noise ratio (SNR) greater than 12 is removed from the sample. A randomly populated parameter space with orbital periods, star limb dimming and radii, planet radii, and orbit inclinations is created. Each light curve with injected planets has been designated '1', while the rest are classified as '0'. These labels are used to train our classifier to recognize the injected transit signals. The training set contains 7,873 light curves, 3,937 belonging to class '1' (planet candidate). This dataset is divided into training-validation sets (90%) and a test set (10%) for 10-fold cross-validation. The next stage is to extract features necessary for encapsulating the light curve characteristics. These features are fed into the classification model during training and inference. TSFresh, a Python framework, is used for efficient feature extraction from time series such as light curves. Metrics include the absolute energy of the time series (sum of squared values), the count of values surpassing the mean, and the coefficients of the one-dimensional discrete Fourier transform. The light curves are resampled to a frequency of 1 hour across all data sets used, guaranteeing consistent distribution of data points across time windows. While not required, resampling is a typical procedure for improving data representation and aiding faster convergence. Resampling entails summing several data points within a certain time and interpolating missing data points on curves. The one-hour frequency chosen balances data loss and avoids considerable data size rise. As a result, these resampled time series can be easily analyzed using various time-series analysis tools. The effective feature extraction setting in TSFresh yields roughly 790 generalized time series features. Following feature extraction, conventional machine learning data pre-processing methods are applied. Irrelevant features, defined as those with constant values across the data set, are deleted, and missing values are imputed using interpolation. Finally, the complete data set is scaled with a robust scaler, which is the foundation for training a tree-based classifier. The classification model is a binary classifier, which divides each light curve into 'planet candidate' and 'non-candidate' classes. The popular machine learning framework lightGBM runs a gradient-boosted tree (GBT) model. The GBT model is an ensemble of progressively trained decision trees. The GBT model is trained in each iteration to duplicate the decision tree findings by fitting the negative gradients, also known as residual errors. During training, a 10-fold cross-validation (CV) is also used. This includes dividing the training set into 10 smaller sets and using 9 for model training at a time, with the remaining set allocated for evaluating model performance. This iterative approach provides thorough examination, with the ultimate performance calculated as the average of computed values. The findings are evaluated using four metrics: accuracy, precision, recall, and the Area Under the Receiver Operating Characteristic (ROC) Curve (AUC). Because of the inherent data set imbalance, accuracy, while often employed, is not an appropriate statistic for evaluating the planet recognition algorithm. While precision ensures precise 'planet candidate' predictions, it may result in overlooking possible planet candidates, making it less valuable. On the other hand, recall, or the true positive rate, is regarded as critical, with identifying genuine planets being prioritized. The decision threshold, a model hyperparameter used for classification decisions, influences the model's performance. This threshold causes the precision-recall trade-off, which influences the categorization outcome. The AUC, which is independent of the decision threshold, enables complete model

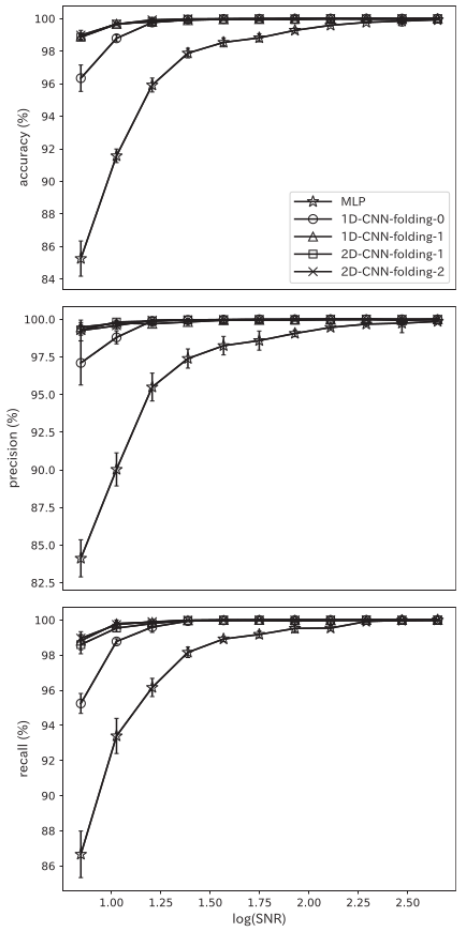


Fig. 2 Accuracy, precision, and recall for all five models as a function of S/N. Model 1 (MLP model) has open stars connected by solid lines; Model 2 (1D-CNN-folding-0 model) has circles with solid lines; Model 3 (1D-CNN-folding-1 model) has triangles with solid lines; Model 4 (2D-CNN-folding-1 model) has squares with solid lines; and Model 5 (2D-CNN-folding-2 model) has crosses with solid lines [6].

evaluation and hyperparameter optimization. AUC and recall are deemed the most important measures in this study, corresponding with the primary goal of maximizing the detection of light curves containing actual planet signals. As a result, hyperparameters are initially optimized to maximize AUC, then the decision threshold is tuned to attain a high recall while keeping as much precision as possible [7].

David J. Armstrong et al. dig into exoplanetary research, focusing on validating exoplanets utilizing novel machine learning algorithms. The rising volume of data from missions such as Kepler necessitates the development of strong and efficient algorithms for distinguishing genuine exoplanetary signals from false positives. The authors begin by setting their work within the larger framework of exoplanetary research, emphasizing the difficulties in finding exoplanets, particularly in the presence of noise and instrumental artifacts. They emphasize the need for precise validation methods in validating the existence and characterizing the attributes of exoplanets. The input data is the Kepler Data Release 25 (DR25), which spans quarters 1-17 and measures stellar brightness for about 200,000 stars over a 4-year period. The NASA Exoplanet Archive provided the data and metadata. The Kepler data is processed using the Kepler data processing pipeline before being detrended by the Presearch Data Conditioning pipeline. The transiting planet search section of the Kepler pipeline identifies planetary candidates, providing Threshold Crossing Events (TCEs) with candidate transits with a significance of $>7.1\sigma$. After passing specific validation procedures called the 'data validation' (DV) process, these TCEs are identified as Kepler Threshold Crossing Event-like Objects of Interest (KOIs). The DV Kepler light curves are used in the same way that the light curves used for the Kepler transiting planet search (TPS) are. Based on the TCE ephemeris, the light-curve data are phase folded and then sorted into 201 equal-width bins in phase, spanning a range of seven transit durations centered on the candidate transit. Following Shallue and Vanderburg (2018), these parameters are chosen for their 'local' view, with a window encompassing one transit period less to improve the resolution of the transit event. Interpolating nearby bins fills empty bins. Following Shallue and Vanderburg (2018), a 'global' perspective is also experimented with, employing 2001 phase bins containing the full phase-folded light curve.

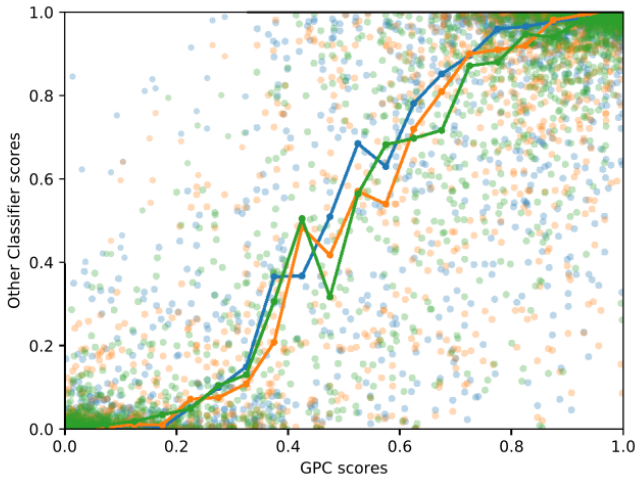


Fig. 3 GPC scores versus RFC (blue), ET (orange), and MLP (green). Overplotted are the median values of 20 evenly distributed bins. The GPC is often more conservative than the other models when creating classifications, resulting in the obvious trend [8].

After scaling, the most significant four feature distributions were evaluated by the optimized random forest classifier (RFC). There are two types of false positives (FPs): astrophysical FPs and non-astrophysical FPs. These sets, when joined and proportions are preserved, produce the FP set for further analysis, which is roughly double the size of the confirmed planet set. The data is divided into three sets: a training set (80%), a validation set for model selection (10%), and a test set for final model performance analysis (10%), with the ratio of planets to FPs kept constant. In addition, for a thorough dataset, a set of TCEs with no disposition is labeled as 'unknown'. There are numerous machine learning algorithms available, each with a particular level of complexity and characteristics. Eight models with various parameters are developed for empirical model selection, testing a total of 822 alternatives. This is accomplished through the use of the SCIKIT-LEARN PYTHON module. The best settings for each algorithm are determined by comparing validation set scores. Most tested models have shown to have excellent success rates. After calibration, the Random Forest Classifier (RFC) performs optimally. Notably, even simpler models such as linear discriminant analysis (LDA) perform well, showing that the underlying decision space is not very complex. To prevent overfitting by any one model, the RFC is chosen to proceed as a versatile and resilient approach, with results augmented by classifications from the next two most successful models, ET and MLP. A Gaussian Process Classifier(GPC) is used for the feature+Light Curve (LC) input data to provide an independent and inherently probabilistic technique for comparison and to prevent overconfidence in model classifications. GPFLOW is used to implement the GPC. Additionally, the GPC is used to test variations of the input data. The findings are determined to be not overly dependent on the input dataset, therefore the feature+LC dataset is utilized to distinguish itself from the other models. Calibration of classifier probability is often accomplished by plotting the 'calibration curve,' which indicates the fraction of class members as a function of classifier output. The usually accepted criteria for statistically validating a candidate as a planet is $p(s = 1|x^*) > 0.99$. To get such an exact probability, the calibration precision must be at least 1% or better. For this aim, the isotonic regression calibration technique is used, which calibrates by counting samples in bins of given classifier scores. At least N = 10,000 test planets are required to minimize the Poisson counting error N/N below 1% to measure the fraction of genuine planets in the $p(s = 1|x^*) > 0.99$ bin. Calibration requires more test inputs due to the size of the training set. To achieve this level of precision, extra planets, and FPs are synthesized from the training set, interpolating between members of each class. A training set member is chosen randomly, followed by another member of the same class who is within the 20th percentile of all member-to-member distances within that class. The Euclidean distance between the values of each column for two class members determines distances. This method of limiting distances allows for non-trivial class borders in the parameter space. Interpolating between the two selected real inputs yields a new synthetic class member. The training set generates a total of 10,000 planets and FPs for calibration, and these synthetic datasets are only utilized for calibration, not training the classifiers [8].

Yucheng Jin et al. present a preliminary investigation into the use of machine learning approaches for the detection of exoplanets. The paper recognizes the growing interest in using machine learning to filter through astronomical data and intends to contribute to this endeavor by undertaking preliminary research. For supervised classification, the Kepler dataset is used. The second dataset (confirmed exoplanets) contains stellar and planetary parameters for confirmed exoplanets. Radius, mass, density, temperature, and other critical characteristics must be considered. This collection contains 4,375 verified exoplanets, some of which were discovered by the Kepler Space Observatory and the remainder by other



Fig. 4 Exoplanet Star Map in the Same Clustering as the Planet Earth [9].

space observatories. For unsupervised clustering, this confirmed exoplanets dataset is used. During Exploratory Data Analysis (EDA), a histogram shows the relationship between a key attribute, the log scale orbital period, and the target variable. The plots demonstrate that confirmed exoplanets have a Gaussian distribution concerning the log scale orbital period in the region of -2 to 4. The candidate is more likely to be a false positive observation if the value of the log scale orbital period is too high or too low. The training approach in this study was divided into two parts. For the supervised learning, the Kepler dataset was used. For the unsupervised clustering, a dataset of confirmed exoplanets was used. During the supervised learning phase, decision trees, random forests, naive Bayes, and neural networks were used. Decision trees were chosen for their capacity to classify data based on information gain while optimizing parameters such as minimal split samples and maximum tree depth. By altering the number of trees, random forests, which aggregate many decision trees, were fine-tuned for optimal accuracy. Assuming feature independence, Nave Bayes provided a probabilistic technique for classification. The activation functions, solvers, layer size, and learning rate of neural networks, commonly used in classification, were optimized. Unsupervised learning used k-means clustering to organize exoplanets and discover those that are similar to Earth, possibly indicating habitable zones. The classification phase exhibited significant accuracies across models, with the multilayer perceptron performing best. Furthermore, cross-validation confirmed that random forest was the most accurate model. Clustering and feature selection were used in the unsupervised learning challenge to discover exoplanets similar to Earth inside specified clusters. The classification phase exhibited high accuracies across models, with the multilayer perceptron model outperforming the others with a 99.79% accuracy. Random forest scored 92.11% accuracy, nave Bayes achieved 88.50% accuracy, and decision tree obtained 99.06% accuracy. On the test set, all four models performed admirably. Furthermore, cross-validation confirmed that random forest was the most accurate model, with an average accuracy of 82.39%. k-means clustering successfully classified exoplanets and found prospective habitable candidates in the unsupervised learning domain, demonstrating the possibility of discovering habitable zones beyond Earth. These findings provide important insights into the application of machine learning in the field of exoplanetary research, emphasizing its potential for efficient and accurate exoplanet categorization and grouping [9].

N. Schanche et al. focus on The emphasis is on Wide Angle Search for Planets (WASP) data, particularly on their robotic devices in La Palma and South Africa. The WASP project, which began with low-cost commercial components, used 8 cameras per location, each with science-grade Charge-Coupled Devices (CCD) for imaging. The WASP field is divided into tiles, each covering approximately a 7.8-degree square field of vision. Data from these tiles was analyzed throughout monitoring seasons to remove systematic inaccuracies. The dataset included over 700 ORCA TAMTFA (ORion transit search Combining All data on a given target with TAMuz and TFA decorrelation) fields, each of which might contain up to 1000 candidates with Box Least Squares (BLS) signals over certain thresholds. Human observers checked in on selected candidates for additional validation. The importance of developing a systematic approach for candidate identification by utilizing the dataset's history classification records for future supervised machine learning attempts is emphasized. Random Forest Classifier (RFC) was chosen to be pursued because of its high recall rate. The RFC training dataset consists of a table of data containing all of the planets (P), eclipsing binaries, both alone and blended with other nearby objects (EB/Blend), variable stars (V), and light curves with no planetary transit after human intervention (X). Blends are extremely common in WASP data due to their 3.5-pixel (48-arcsec) aperture, which causes signal blending from several stars. Additionally, additional features are included to capture more abstract and relational information, such as the transit depth-to-breadth ratio and the magnitude distribution skewness inside the transit event. The skewness can discriminate between shallow 'V' shaped eclipses of grazing stellar binaries and 'U' shaped center transits of a tiny planet across a bigger star. To calculate the difference between the estimated main sequence radius and the measured radius, precise distance computations from Gaia data are used. The dataset currently has 34 features in total. The entire dataset, including star names, descriptive attributes, and disposition, is randomly divided into a training dataset of 4,697 cases and a test dataset of 2,314 samples prior to training. To improve classifier performance, all features in the training dataset are median-centered and scaled, with the scaling parameters retained for future classification datasets, including the testing dataset. The training dataset is subjected to a variety of classifiers, including Support Vector Classifier (SVC), Linear Support Vector Classifier (LinearSVC), Logistic Regression, K-nearest Neighbours (KNN), and Random Forest Classifier (RFC). Each classifier is parameter-tuned using a grid of combinations particular to that classification method, resulting in the best settings. While certain classifiers, most notably the K-nearest neighbor classifier, perform poorly due to the underrepresentation of the class of interest, planets, we solve this imbalance using the Synthetic Minority Over-sampling Technique (SMOTE).SMOTE provides synthetic data points for minority classes, which improves planet retrieval from the dataset. Despite a minor rise in erroneous positives for planets, SMOTE significantly reduces false positives for the Support Vector Classifier (SVC) while keeping consistent levels of true positives and true negatives. The Random Forest Classifier function from Scikit-learn was used for classification and parameter adjustment. The ideal forest consisted of 200 trees, each with a depth of 6 and 6 random features at each split to reduce overfitting. The RFC technique has the advantage of including a wide range of characteristics, not all of which must be effective predictors. This enables inclusive exploratory data analysis while avoiding biases introduced by input list curation. However, it is critical to recognize that the RFC's performance depends on the quality of input features, implying that better feature identification could lead to improved performance. The RFC performed an analysis during the training process to identify the relevance of various characteristics in prediction. This analysis sheds light on the classifier's decision-making process, allowing for future investigation. For example, due to the incidence of false planet detections linked with diurnal systematics and orbital periods near to multiples of one sidereal day, the planet's period emerged as a key indicator. In contrast, stellar parameters such as magnitude and radius of the star had little impact on the overall forecast, indicating that there was no clear preference for a specific star size to host a specific item in the sample. Skewness was the lowest-ranked feature, signifying poor capture of transit shape information. The RFC results showed a recovery rate of roughly 93% for confirmed planets within the dataset, with the misclassification of more than 10% of EB/Blends as planets being of particular concern. Because of the high false positive rate, Convolutional Neural Networks were created to act as a secondary test on the light curves. Although each WASP field contains a different amount of sample points, the input to the CNN for all samples must be the same length. The folded light curves are separated into 500 bins in transit-phase space to standardize the data length. The number of bins was found empirically, with 500 being the best compromise between presenting comprehensive light curves and many bins missing data. The neural network was implemented using the Keras package, with multiple activation functions and weight initialization strategies. For intermediate layers, the neural network used the rectified linear unit (relu) function, and for the output layer, it used the sigmoid function. The Adamax optimizer, an iterative approach that optimizes gradient descent, was used to oversee the learning process. The CNN was designed for direct usage of light curve data and included convolution and pooling stages, as well as dropout layers to prevent overfitting. To overcome the imbalance in the training set, more training instances were constructed by adding artificial transit patterns into existing WASP light curves that lacked such signals at the outset. This enhancement entailed injecting synthetic transit patterns to ensure a realistic depiction of normal noise patterns and correlated and uncorrelated noise. Light curves indicating the absence of observable planet signals, were chosen specifically. Those with the least variance below one of the best-fit data were picked, yielding a total of 848 light curves. The CNN included two convolutional

layers, pooling stages, and dropout mechanisms. Using local transit information enhanced performance. The performance of the CNN was greatly impacted by missing data handling strategies, with the best results obtained by adding a specified placeholder value into the whole light curve and repeating the last valid value in the local light curve. When analyzing the test dataset with binned light curve input, the CNN achieved an overall accuracy of roughly 82%. The CNN beat the RFC in finding eclipsing binaries and blends, indicating greater efficiency for follow-up analysis [10].

Megan Ansdell et. al focuses on the use of deep learning to improve the classification of exoplanet transit signals using space-based observatory data, notably the Kepler mission and, soon, the Transiting Exoplanet Survey Satellite (TESS). The study extends the work of Shallue and Vanderburg (2018) by embedding scientific domain information into the deep learning architecture, resulted in a considerable increase in model performance. The study employs a deep learning model known as Exonet, which expands on earlier research by including new variables such as centroid time series and critical stellar properties. The incorporation of these features significantly improves the accuracy and precision of transit classification. The Exonet model has an amazing 97.5% accuracy and an average precision of 98.0%. The significant increases in recall for low signal-to-noise transit signals, which could correlate to rocky planets in the habitable zone, are especially interesting. This boost in recall is critical for detecting weak signals in noisy data. The addition of scientific domain knowledge, such as centroid time series data and stellar properties, is critical for improving the model's overall performance. By exploiting this domain expertise, the model is better able to detect false positives such as background eclipsing binaries (BEBs) and other instrumental noise artifacts, resulting in increased precision and recall. Furthermore, the work uses data augmentation strategies to address model overfitting, which is a typical problem in machine learning models. By supplementing the training data, the model becomes more resilient to fluctuations and uncertainties in the input data, resulting in better generalization performance. Overall, integrating these scientific domain knowledge and data augmentation techniques improves the model's accuracy and precision while also emphasizing the importance of expert knowledge in improving the capabilities of cutting-edge deep learning models for scientific research applications. The study not only introduces the Exonet model but also compares its performance to the smaller, more compact Exonet-XS. Notably, Exonet-XS maintains better model performance while drastically lowering its size. The research also looks into precision-recall curves, providing a thorough evaluation of model performance using various thresholds and signal-to-noise measurements. Furthermore, the application of deep learning in transit categorization is highlighted as being especially relevant for prospective space-based photometry missions aimed at discovering tiny planets, such as TESS and PLATO. The study's findings, methodology, and model architecture are valuable to the field of astrophysics, and they considerably enhance the use of deep learning to identify and classify planetary transit signals. The research recognizes NASA and NSF grants and makes the code and data used in the study available to the public [11].

Kai Hou Yip et al. generated 3 synthetic planetary atmospheres from planets on the Ariel Target list (Edwards et al. 2019a). A total of 11,940 transmission spectra were generated. The transmission spectrum measures the λ dependency change in transit depth (Δt_λ). The Alfnoor-forward function in Alfnoor (Changeat et al. 2020a) and the Ariel Radiometric Model (Mugnai et al. 2020) allow for large-scale spectrum creation. Each spectrum is binned to Ariel Tier-2 resolution, with error bars estimated based on Deep survey requirements and reasonable instrument, observation, and planet estimations. The Ariel Tier-2 resolution remains constant throughout the inquiry, with no wavelength-based binning. The article discusses the setup used by Changeat et al. in their investigation and provides a benchmark for comparison. The binned spectrum is denoted as the mean ground-truth spectrum, with $\{x_i\}$ representing transit depth. The atmosphere is assumed to have a constant He/H2 ratio of 0.17, with Rayleigh scattering and Collision Induced Absorption included. Other planetary parameters such as stellar radius, planet radius, mass, temperature, and orbital parameters are taken from predictions. Trace gases are added to generate an unbiased sample of spectra, with their log abundance uniformly sampled. Gray clouds with a cloud deck pressure log are also included. To differentiate across contexts, Atmospheric Model Parameters are referred to as "AMPs" and Deep Neural Network parameters (synaptic weights) as "weights". To train DNN models, the synthetic spectra and corresponding AMPs are normalized to have zero mean and unit variance. The standardized data set is randomly divided into three subsets: original training (70%), validation (10%), and test (20%). The original training set is not directly used during training. Rather, it is utilized to create an expanded training dataset. The enlarged training set includes 50 data points for each spectrum (X) in the original training set. Each X is generated by sampling a new x_i from a Gaussian distribution centered at x_i, with a standard deviation specified by the associated σ_i. The ground-truth (noise-free) spectra are eliminated, and models are trained on noisy, more realistic examples. The validation set contains noise, but the test set is clear of it. A Deep Neural Network (DNN) to handle multiple output regression tasks. To facilitate referencing, the AMPs are denoted as $y = [y_1, y_2, ...y_9]$, with y_j being the jth AMP (as arranged above). The model is trained in a supervised way by minimizing the Mean Squared Error (MSE) between projected values $\hat{y}$ and ground-truth values y, averaged over all targets. Three DNN architectures (MLP, CNN, and LSTM) were compared for inferring atmospheric model parameters from exoplanet spectra using simulated Ariel data. Approaches are provided for analyzing regression model performance, detecting sources of error through bias and variance, and assessing forecast confidence. Using these evaluation approaches, it was discovered that the three DNN models we trained behaved

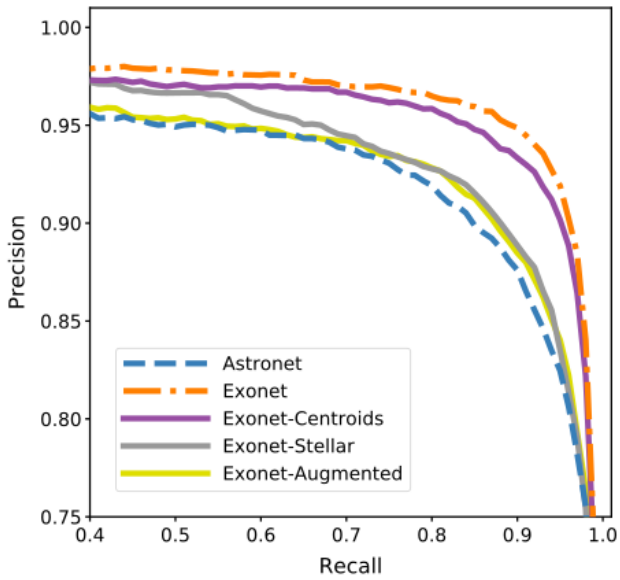


Fig. 5 Astronet's precision-recall curves were compared to those of Exonet with various additions of scientific domain information to demonstrate the specific contributions to gains in model performance. Exonet-Centroids just adds centroid curves, Exonet-Stellar adds stellar parameters, and Exonet-Augmented adds our supplemental data augmentations. Exonet then integrates all of these enhancements into a single model [11].

similarly and could accurately determine molecule abundances down to $10^{-5.8}$. A perturbation-based sensitivity analysis was developed to evaluate the impact of each feature (wavelength bin) on predicting each goal (atmospheric parameter) for a trained model. This study confirms that the DNN models' predictions closely match the physical intuition for atmospheric parameter spectral signatures, Ariel instrument specification, and Ariel Deep survey observational technique [12].

The proposed paper improves classifier performance by including transit synthetic light curves based on expert knowledge of the planet's transit patterns. Mandel and Agol established the quadratic model for limb darkening rules, which is used to get these results. The flux f during transit over a star disc with quadratic limb darkening is:

$$f = 1 - \frac{(1-c)\lambda_e(k,z) + c\lambda_d(k,z) + \mu_2\eta_d(k,z)}{1 - \frac{\mu_1}{3} - \frac{\mu_2}{6}}$$

In this equation, k is the radius ratio, z is the projected distance, $c = u_1 + 2u_2$, u_1 and u_2 are the quadratic limb darkening coefficients, and λ_e, λ_d, and η_e are k and z-dependent functions. This model uses the PyTransit Python package and includes parameters such as transit duration τ, planet radius to stellar radius ratio (r_p/r_s), orbital semimajor axis to stellar radius ratio (a/r_s), and orbit inclination (i). The light curves are pre-processed to remove any empty intervals. The missing values are replaced with the average of the empty interval's neighborhood. The noise of less than 10% magnitude is introduced to these new numbers to improve their compatibility with reality. The light curves were interpolated to 4000 points, which equals 10 periods of 400 points each. Finally, the data is normalized so that all values are between 0 and 1. The phase folding technique is introduced to make use of transit periodicity. It entails folding each light curve on the transit period in the catalog and binning it to produce a 1D vector of an improved signal. This strategy improves transit detection, but the transit period must be known in advance; otherwise, the folding period may change, rendering the transit invisible to the model. Finding transits in newly published observational data might be challenging due to the aforementioned circumstances. To address this issue, a separate phase folding method is used to build a 2D representation by folding each light curve on a period that may differ from the transit period, increasing transit detection independent of the transit and folding periods. Following this procedure, the detection model inputs were created by folding the light curve into ten segments and using the transit period from the catalog's koi_period column. The classification objective is to categorize each light curve into two categories: planet candidate and false positive. This application makes use of a variety of deep-learning algorithms. In our method, a convolutional neural network was chosen because it outperforms artificial neural networks in classification tasks involving spatially aligned data, such as images or sounds. Because CNN takes advantage of the spatial structure of the output to detect local features that only need to be learned once, the number of trainable parameters, memory usage, and desired output computations will all be reduced. Transfer learning involves taking a big network that has already been trained for a certain problem and applying it to a new problem. This adjustment is made at the network's end, changing the number of output neurons to match the number of classes in the new problem (2 classes). This is a particularly valuable technique because the earliest stages of the network typically identify broad features that may be applied to almost any classification task. Training

is required to adjust the last layer to the new classes, but transfer learning eliminates the need to retrain the entire network. You can choose which layers to train or not train. Given the computational cost of training such a large network, this is quite efficient. The suggested convolutional neural network architecture is based on Xception, created by Francois Chollet. This CNN acquired rich feature representations from the ImageNet dataset, outperforming Inception V3 on it (for which it was built) and greatly outperforming Inception V3 on a bigger image classification dataset of 350 million images and 17,000 classes. Xception is a linear stack of 71 depth-wise separable convolution layers with residual connections. It can classify 1000 different types of objects and has a picture input size of 299x299. DCNN models were generated with $R = 483$ real curves and a ratio of synthetic curves S with transit defined by the λ parameter to assess the impact of increasing synthetic data on training. To keep the training set balanced, the same amount $S + R$ of non-transit curves are introduced. The basic idea of this work is that the model's performance can be improved by utilizing a ratio λ of synthetic light curves during the training stage. The detection performance was tested on a test set of $R = 100$ real curves with transit and the same number of real curves without transit, under two extreme scenarios - Training using only real light curves with transit ($S = 0$, $R = 483$) and $S + R$ real light curves without transit, Training using just synthetic light curves with transit ($S = 483$, $R = 0$) and $S + R$ real light curves without transit. The neural network threshold is fixed at $T = 0.5$ in both circumstances since this is a binary classification problem with 0 representing a predicted light curve without transit and 1 representing a predicted planet candidate. The model trained with only real curves can recognize 74% of transit light curves from the test set with a precision of 74.7%, which is a good rate but falls short of the previously published metrics. On the other hand, when the model is taught with synthetic curves, it can recognize transit light curves with comparable precision to the model trained with real curves, but it lacks the variability of transit curves. As a result, training with real curves adds diversity to the model's identification of transit light curves, whereas incorporating synthetic light curves improves prediction precision. To determine the optimal λ and T ratio, coarse tuning with heuristic optimization and sensibility analysis is suggested by the authors [13].

Shay Zucker and Raja Giryes trained the network on simulated data and tested its performance. They constructed 10 5 pure noise light curves, with or without a transit signal, totaling 2×10 5 light curves. The neural network was trained using the first 2×83333 light curves before testing it on the remaining 2×16667. In the simulation, a fictitious planet-detection space telescope was used with a five-minute cadence, which falls between Kepler's long-cadence rate of 30 minutes (Borucki et al. 2010) and PLATO's anticipated 25-second cadence (Rauer et al. 2014). For the sake of simplicity, the mission's observing run was supposed to last 21.33 days or 6144 samples. The expedition emulated a stellar population that included stars with apparent magnitudes ranging from $V = 10$ to $V = 16$. These magnitudes were created using a beta distribution with coefficients $\alpha = 3$ and $\beta = 1$, followed by a linear transformation to fit inside the required interval [10, 16], resulting in a distribution biased towards fainter magnitudes. The impact of magnitude on uncorrelated photon noise is determined by the features of the observational apparatus. A Gaussian Process (GP) technique was used to add correlated (red) noise, with a kernel consisting of squared-exponential and quasi-periodic components (Aigrain et al. 2016). Each red-noise light curve in the dataset was included twice: once in its original form and again with an extra transit signal. Because this is a feasibility study, only transits lasting less than 6100 minutes (approximately four days) were simulated, assuring at least five transit events during the observation run. The simulated transits were modeled as basic trapezoids, except the complete comprehensive model, which included limb darkening (Mandel and Agol 2002). Transits were assumed to be strictly periodic to minimize problems caused by many planets and transit-timing variations (TTV). The transit phase and combined ingress-egress relative duration (as a fraction of transit duration) were picked at random from uniform distributions with values ranging from 0.01 to 0.3. The CNN used consists of four convolutional layers, all with a stride of 4. Following deep learning terminology, this is comparable to performing a standard convolution, then sampling every fourth entry and discarding the remainder. The first layer used 40 separate convolution kernels of length 40, resulting in a two-dimensional signal of size 1527×40. The second layer had 32 kernels of size 23×40, with each convolution receiving input from all 40 channels, totaling 32 channels. The third and fourth levels featured 32 kernels measuring 23×32. The kernel sizes indicate a trade-off between minimizing the number of parameters and increasing the size of each "neuron's receptive field." Each convolution is followed by ReLU nonlinearity (Nair and Hinton 2010) and batch normalization (BN) (Ioffe and Szegedy 2015). The ReLU is an element-wise function that zeros off any negative values in its input, thereby projecting onto the positive orthant. The BN operation normalizes a particular layer's output across different data samples, resulting in a zero mean and unit variance. ReLU and BN introduce nonlinearity into the network, which is an important element of deep learning models. The network ends with a 544×2 fully linked layer, followed by ReLU and softmax. The softmax is typically employed after the network and emphasizes the greatest value in the input by applying an exponent element-by-element followed by a division by the sum of the values. The softmax's output can be seen as a probability distribution, showing whether the signal is a noisy transit signal or pure (red) noise. The detection method is one-hot encoding. For optimization, the standard stochastic gradient descent (SGD) method with mini-batches is used. Unlike normal gradient descent, SGD computes the gradient at each iteration using a random subset of the data. The training phase lasts 35 epochs, during which the network converges.

Training for a few extra epochs did not result in noticeably different results. The ADAM optimizer (Kingma and Ba 2014) is used, which adjusts the learning rate based on predictions of the first and second moments of the gradients [14].

The exploration of space missions, as demonstrated by NASA's Transiting Exoplanet Survey Satellite (TESS), has ushered in an era of unprecedented data collection, notably in the field of photometric observations. Working in combination with its predecessor, Kepler, TESS has considerably increased our understanding of exoplanets, to double the number of known exoplanets. To deal with the massive job of quickly analyzing TESS's massive photometric data, astronomers pioneered automated triage and vetting procedures. These methods use deep learning models, demonstrating excellent precision and accuracy in discriminating planet candidates from other signals while also addressing the time-consuming and occasionally inconsistent nature of human screening processes. Machine learning techniques, specifically convolutional neural networks (CNNs), have emerged as effective tools for categorizing TESS candidates. Researchers have consistently improved old models and proposed novel CNN architectures that are customized to the specific properties of TESS and Kepler data. Despite initial hurdles in adopting the AstroNet model for TESS candidates, recent advances have been made, including the creation of a CNN trained on genuine TESS data. This model has shown encouraging results in both triage and vetting tasks, demonstrating its ability to recognize celestial entities using transit-like signals. The study methodically investigates the complexities of dataset compilation, relying on TESS light curves from specific sectors and focused on threshold crossing events (TCEs) detected using the MIT Quick Look Pipeline. The pipeline uses catalog-based circular aperture photometry, detrending techniques, and transit searches with the Box Least Squares algorithm. Ground truth labels for TESS TCEs fall into several categories, including planet candidates, eclipsing binaries, star variability, and instrumentation noise. The study also emphasizes the need for homogeneous labeling to maintain dataset consistency. A thorough examination of the neural network architecture reveals particular changes for triage and vetting modes, as well as the addition of a secondary eclipse view as a potential model improvement. Training details, such as model averaging for robust predictions, highlight the efforts to reduce stochastic variations between distinct models. The neural network's performance is assessed using metrics such as precision, recall, and average precision. The models, notably AstroNet Triage, are highly accurate, with a precision of 0.749 and a recall of 0.975 on a test set of 1650 TCEs. This methodology effectively eliminates non-planetary signals, recovering all 49 planet candidates and the vast majority of eclipsing binaries while removing 93% of the negatives. Certain planet candidates are difficult to differentiate from eclipsing binaries, resulting in false positives. The study emphasizes the complementarity of machine and human vetting by offering impartial attitudes and solving the challenges that neural networks confront when detecting anomalous astrophysical signals. The narrative continues with the application of the established models to TESS Sector 6 data, emphasizing the effective reduction of false positives as well as considerations for the sector's various star populations. The study continues by discussing potential routes for model improvement, the need for precisely generated data for training, and the model's generalizability to previously unexplored areas. It also proposes future study initiatives, such as adding more features and investigating follow-up observations to corroborate the planetary nature of observed threshold-crossing events [15].

The abundance of data provided by the Kepler and Transiting Exoplanet Survey Satellite (TESS) missions, which includes over 100,000 potential transit signals, has sparked an increased interest in using machine-learning approaches to identify new exoplanets. In this context, ExoMiner, a unique deep learning classifier suggested in this paper, deviates from traditional methods by simulating the diagnostic inspection performed by domain experts to validate transit signals. ExoMiner, known for its high accuracy, explainability, and resilience, is used to validate 301 new exoplanets from the MAST Kepler Archive and exhibits relevance to missions like as TESS. The extensive experimental examination demonstrates ExoMiner's higher reliability and accuracy when compared to existing transit signal classifiers, as indicated by numerous classification and ranking criteria. For example, with a fixed accuracy value of 99%, ExoMiner achieves 93.6% recall, which is higher than the best successful existing classifier's 76.3% recall. Notably, ExoMiner's modular structure improves its explainability, aided by the inclusion of a simple framework. This framework provides specialists with insight into the reasoning underlying ExoMiner's classification of transit signals into specific class labels, such as planet candidate or non-planet candidate. ExoMiner's inclusion into the field of exoplanet finding, thanks to its original approach and solid performance, helps to drive the continued advancement of machine learning applications in transit signal analysis [16].

In recent years, there has been a surge in the application of machine learning algorithms in the field of astronomy, revealing their potential across a wide range of jobs. The identification of exoplanets and related phenomena such as transits, which are critical for understanding celestial bodies outside our solar system, has been a major emphasis. The Random Forest classifier and Convolutional Neural Networks (CNN) emerged as the most effective ways of recognizing planet-like transits among the many machine learning algorithms studied. The Random Forest classifier shows great accuracy in transit data analysis due to its ability to handle complicated characteristics and patterns rapidly. Similarly, CNN, which is well-known for its image recognition abilities, was extremely good at finding transit signals within astronomical data, which proved especially useful in exoplanetary transit research. Furthermore, the Box Least Squares (BLS) technique was recognized for its usefulness in detecting periodic transit-like signals, highlighting its importance in exoplanetary transit research. Among other applications, this method demonstrated

its capacity to detect exoplanetary transits and periodic fluctuations in observed data. It's worth noting, however, that other machine learning models, such as Support Vector Machines (SVM) and Decision Trees, were examined while acknowledging their shortcomings. SVM and Decision Trees have been observed to have difficulties addressing sophisticated feature interactions and may be limited in their usefulness inside specified data sets. The efficient utilization of machine learning approaches offers tremendous promise in the broader context of the exoplanetary study. Using these advanced methodologies not only improves our understanding of exoplanets and transit occurrences but also offers up possibilities for discoveries in the field of astrophysics. As the search for exoplanets progresses, incorporating machine learning into astronomical data analysis remains an important and active area of research and discovery.

III. Materials and Methods

A. Kepler dataset

NASA's Kepler space telescope mission produced the Kepler dataset, which has become a cornerstone in exoplanetary study. Kepler, launched in 2009, watched a large section of the Milky Way, studying roughly 200,000 stars in an attempt to locate exoplanets using the transit method. The spacecraft precisely captured the brightness variations of these stars, and the collection includes a large archive of light curves reflecting these variations across time. Kepler's rigorous observations aided in the discovery of hundreds of exoplanets, encompassing a diverse range of planetary systems ranging from hot Jupiters to Earth-sized planets. This dataset is large, containing high-fidelity photometric time series data that reveal the recurring dimming of stars, which indicates the presence of transiting exoplanets. This massive dataset has enabled researchers to identify planetary candidates, determine their diameters and orbital periods, and, in some cases, estimate their atmosphere qualities. During the mission's existence, Kepler identified thousands of exoplanetary candidates and validated hundreds of them, revolutionizing our understanding of the diversity and features of exoplanetary systems.

B. Transiting Exoplanet Survey Satellite (TESS)

The Transiting Exoplanet Survey Satellite (TESS) dataset is a priceless resource for exoplanetary researchers today. TESS, launched by NASA in 2018, focuses on identifying exoplanets via the transit method. TESS, in contrast to Kepler, surveys the entire sky by dividing it into 26 sectors and methodically monitoring each sector for nearly a month. This extensive survey method has resulted in the finding of numerous exoplanets, including a considerable number of tiny rocky planets and fascinating systems. TESS data is a goldmine of precise photometric data, documenting fluctuations in star brightness that indicate prospective transits. The dataset is critical for characterizing exoplanets, investigating star behavior, and expanding our understanding of the diversity and features of exoplanetary systems. Ultimately, we hope to gain a better understanding of the universe's immense cosmic panorama.

C. Wide Angle Search for Planets (WASP)

The Wide Angle Search for Planets (WASP) dataset is a critical resource in exoplanetary research, especially for ground-based wide-field transit studies. It comprises data collected by two robotic sensors in La Palma and South Africa. For imaging, the WASP project employs an array of eight commercial cameras outfitted with science-grade Charge-Coupled Devices (CCDs) that are strategically positioned to cover a large area of the sky. Observations of various celestial objects in specified sky tiles are included in the collection. To reduce systematic mistakes, the gathered data is meticulously processed, including detrending and decorrelation. The generated light curves are meticulously analyzed to find and characterize extraterrestrial transmit signals.

D. Random forest classifier

The Random Forest Classifier (RFC) is a machine learning ensemble approach that is particularly useful due to its ease of implementation and, consistent performance across various applications. Because of the RFC's inherent advantages, particularly its feature ranking traceability, multiple techniques used to search exoplanets favor RFC over linear Support Vector Classifier (SVC), or K-Nearest Neighbour (KNN) approaches. The RFC comprises many independent decision trees, each of which is trained on a random part of the training dataset and uses randomized feature subsets to make splits that optimize correct classifications. Combining these trees results in a more robust and generalized predictor than the individual trees. RFC can identify fresh light curves after training, assisting in the automated identification of probable exoplanetary candidates. It can tell the difference between different transit shapes, durations, and depths, allowing it to discern different types of transiting events and probable false positives. The RFC results show that for confirmed planets in the collection, the recovery rate is around 93%. However, misclassifying more than 10% of eclipsing binaries and blends as planets raise concerns, indicating the necessity for careful evaluation and verification in follow-up observations to minimize resource waste.

E. Gaussian Process Classifier

The Gaussian Process Classifier (GPC) is a machine learning model that is mostly used in the categorization and detection of exoplanets. GPC is based on Gaussian process concepts, essentially probability distributions over functions. It estimates the likelihood that a given data point belongs to a specific class, making it an effective tool for classification jobs where uncertainty assessment is critical. A dataset of light curves can be used to train GPC, which will learn the traits associated with planetary transits and their changes. Once trained, GPC can recognize probable exoplanetary transits and classify new light curves. It can also effectively manage noisy or partial observations, making it appropriate for the inherently unpredictable character of astronomical data. It can help distinguish between different sorts of transiting occurrences,

such as planetary transits, eclipsing binaries, or false positives, improving the classification process's efficiency and accuracy.

F. Convolutional Neural Networks

A CNN is a specialized artificial neural network architecture developed by LeCun et al. in 1990 and popularised by Krizhevsky et al. in 2012. Convolutional Neural Networks (CNNs) have proven to be extremely useful in the hunt for exoplanets, especially when analyzing light curve data. Because of its capacity to learn and extract detailed characteristics from time-series data, the CNN architecture is well-suited for exoplanetary research tasks. Convolutional layers add filters to the light curve to discover important patterns at various scales. The information is subsequently downsampled via pooling layers, maintaining only the most important elements. A CNN is trained by feeding it a labelled dataset with the presence or absence of exoplanetary transits. Through an iterative optimization procedure, the network learns to distinguish between these patterns. Once trained, the CNN can analyze fresh light curves and classify them based on previously learned patterns, allowing it to detect probable exoplanetary signals.

G. k-means clustering

K-means clustering is an unsupervised machine-learning technique used in exoplanetary research, namely in the study of light curves collected from telescopic observations. This approach divides light curves, representing a star's brightness over time, into discrete groups depending on their transit properties. Transits, dimming patterns created by exoplanets passing in front of their home stars, are quite interesting. These light curves can be classified into clusters with comparable transit characteristics using K-means clustering. This grouping makes automated categorization and preliminary evaluation of prospective exoplanetary candidates easier. It is an important stage in the initial screening process since it narrows the group of candidates for deeper scrutiny, making future analysis and validation more efficient.

H. Multilayer Percepton

Multilayer Perceptrons, a type of artificial neural network, are widely used in exoplanetary research due to their capacity to simulate complicated interactions within data. These networks comprise interconnected layers of neurons, each applying an activation function to its input. MLPs are used in exoplanetary research for a variety of tasks, including the identification and classification of exoplanetary transits in lightcurve data. MLPs learn to recognize complicated patterns and features in lightcurves by altering the weights of connections between neurons during training. This enables MLPs to differentiate between various types of transits, false positives, and other astrophysical events. MLPs are an effective tool in automated validation and classification processes, increasing the efficiency and accuracy of discovering probable exoplanetary signals within a given dataset like Kepler or TESS.

I. Gradient boosted trees (LightGBM)

Gradient Boosted Trees, of which LightGBM is an example, are gaining popularity in exoplanetary research due to their efficiency and effectiveness in dealing with complex datasets. For classification problems, LightGBM, an ensemble learning method, employs an ensemble of decision trees. It is coveted for its processing speed and capacity to capture complicated feature interactions in data. LightGBM is utilized in studying exoplanets for tasks such as categorizing exoplanetary candidates based on a profusion of varied properties retrieved from light curves. Its ability to handle enormous amounts of data and capture complex linkages makes it an effective tool for automated candidate vetting and prioritization.

J. Locality Preserving Projections

Thompson et al. proposed a new measure for detecting transiting exoplanet signals that use Locality Preserving Projections (LPP) for dimensionality reduction and k-nearest neighbors to detect transit-like structures within light curves. LPP substantially reduces data dimensionality, displaying tolerance to outliers compared to approaches such as Principal Component Analysis (PCA), and supports transit-like signal grouping by calculating the Euclidean distance between each data point and its k-nearest neighbors. The proposed metric is the mean of the k distances between each neighbor. A Transit-like Conditional Event (TCE) is classified as a transit if the LPP metric shows proximity to other transit-like events in a dimensionality-reduced space of 20 dimensions produced using LPP algorithm.

K. Box Least Squares

The Box Least Squares (BLS) algorithm is used in exoplanetary research to detect transiting exoplanets by analyzing starlight curves. This algorithm iteratively investigates a set of trial periods based on expected planetary behavior, folding the lightcurve data as it goes. A box function describing the transit shape is convolved with the folded light curve for each trial period, and a detection statistic based on the goodness-of-fit is calculated. Peaks in this statistic indicate probable transit events, which aid in identifying exoplanetary candidates.

IV. Result Analysis

In the mentioned paper, except for the CNN, each model underwent 100 training and testing iterations, encompassing all possible input configurations, including binned inputs without MRA and Discrete Wavelet Transform (DWT) cAs (Approximation Coefficients) or cDs (Detail Coefficients) throughout six decomposition levels for each wavelet. To save lengthy training periods, the CNN was trained once and then tested 100 times. Across all model executions, the scikit-learn python API train test split function was used to divide and shuffle the dataset into 60% for training and 40% for testing. Over 100 iterations, the average accuracy, precision, recall, specificity, and execution time for each input configuration were measured. These metrics evaluate the performance of ML models based on the right categorization of "yes" and "no" answers

(True Positives, TPs) as well as incorrect classifications (False Positives, FPs, and False Negatives, FNs). The best model results are highlighted for each input combination, with or without DWT. The models were run on a computer with certain hardware and software requirements. Hypothesis tests, notably Welch's t-tests, were used to statistically establish the superiority of DWT results over binned light curves without MRA. The t-test findings revealed significant differences in most cases, indicating DWT's increased performance, particularly in execution time. Notably, the DWT reduced execution time significantly without influencing other performance indicators. Even without DWT inputs, different ML models performed differently, demonstrating the importance of model architecture. On the 3-median dataset, the CNN with DWT got the greatest accuracy and significant execution time reduction, indicating that MRA improved its performance. Furthermore, the Random Forest model improved its performance with DWT, despite previously producing the greatest results without DWT. Finally, reducing the number of bins without DWT revealed a trade-off between model performance and execution time, highlighting the importance of extracting relevant features from light curves via DWT coefficients while still requiring the binning process for fixed-length inputs to ML models [2].

The DR24 Kepler Robovetter uses the LPP transits measure to determine if a Threshold Crossing Event (TCE) qualifies as a planetary candidate. The Robovetter is critical in determining a pass/fail criterion. Historically, efforts have leaned toward maintaining potentially suspect transits rather than discarding them, potentially resulting in more false positives. Even when implemented cautiously, the LPP metric effectively dramatically lowers false alarms. It specializes in detecting TCEs that look like sine waves rather than genuine transits, especially those with small periods and extended durations. However, there is an acknowledged difficulty when evaluating transits on variable stars, particularly when the variability coincides with a harmonic of the transit period. In such circumstances, the detrending procedure may distort the signal, misclassifying it as a transit. This problem was solved by providing two detrending alternatives during the vetting process, which helped preserve the transit signal when one detrender may distort it. To improve accuracy, the idea offers embedding intelligence into the Kepler Robovetter to aid in selecting the most optimal detrender [5].

The 2D-CNN-folding-2 model demonstrated remarkable performance. The dataset created contained 50,000 light curves, 40,000 labeled as training data, and 10,000 as validation data. The folding method used a period difference percentage (Df) of 1%, resulting in 120 folding periods that were evenly distributed between [0.99σ, 1.01σ]. The analysis tried transit identification by folding each light curve with 120 different periods and leveraging the 2D-CNN-folding-2 model. The results for four different light curves were presented, confirming the model's capacity to detect transits and determine transit times with little uncertainty. The recall, which represents the percentage of recovered transits, was plotted as a function of log10 ($r_p r_s$ in the following analysis. When compared to Christiansen et al. (2016), the model performed better, especially for larger planet sizes ($r_p/r_s > 0.03$). However, due to a lack of training, the model's performance was limited for smaller planet sizes ($r_p/r_s < 0.01$). The study acknowledged the significance of neglecting smaller planets while highlighting the model's efficiency within its well-trained environment. The importance of folding in retaining high accuracy even with low signal-to-noise ratios (SNR) was demonstrated. The durability of the 2D-CNN-folding-2 model was demonstrated even with a 20% variation between folding and transit timeframes, demonstrating its promise in transit detection in new data releases [6].

The Gradient Boosted Tree (GBT) model exhibited versatility to various data sources, including Kepler, K2, and TESS, allowing the development of a global classifier capable of directly processing light curves of varied durations from any source. This adaptability allows for model utilization to be maintained even when data formats change, as demonstrated by the shift encountered with K2 due to temporal gaps in light curves. Notably, the model's performance in the TESS dataset differed from that in Kepler and simulated data. The key contributing factor was a significant class imbalance in TESS data, with less than 3% of light curves containing transit signals, most of which were single transits. Addressing this mismatch with a refined dataset with a more fair class distribution from the ever-expanding number of confirmed exoplanets in TESS data can improve performance, potentially matching Kepler and simulated data levels. A shift to a semi-supervised methodology was considered when considering alternate approaches, departing from the traditional supervised learning path. This proposed method requires embedding data in a lower-dimensional space, maybe using techniques such as autoencoders or principal component analysis. Following that, unsupervised clustering algorithms can discover data clusters, providing a foundation for limited manual labeling using well-understood light curves. Though left for future investigation, this option has the potential for additional model refining. This technique consists of three main steps: preprocessing raw light curves to reduce systematic effects and variability, feature extraction using the TSFresh library, and using these features to build a gradient-boosted tree model. Compared to BLS, the model demonstrated greater accuracy in recognizing light curves with planet signals, significantly lowering false positives. While pioneering work in this sector has investigated deep learning approaches, our study deliberately strayed from this path to pursue a lightweight model approach. The three-step process exhibited efficiency, eliminating the requirement for feature manual selection and specialized equipment such as GPUs. The model's ability to distinguish between crucial and non-critical features improved its effectiveness. Performance studies revealed that the model outperformed BLS at recognizing planet signals, indicating a high potential for

ML model	Accuracy (%)	Precision (%)	Specificity (%)	Recall (%)	Time (seconds)
Sigmoid MLP(5, 2)					
Inputs without DWT.	84.64	77.43	76.31	92.79	12.42
Inputs using coif5 cDs ($j = 4$).	94.23	92.38	91.69	96.72	1.76
Relu MLP(5, 2)					
Inputs without DWT.	49.98	42.67	15.0	85.0	7.65
Inputs using bior2.4 cDs ($j = 3$).	95.12	92.94	92.27	97.93	2.54
Sigmoid MLP(1024)					
Inputs without DWT.	86.04	79.85	72.87	98.93	166.42
Inputs using bior2.4 cAs ($J = 6$).	92.65	89.72	88.13	97.1	46.94
Relu MLP(1024)					
Inputs without DWT.	93.31	88.95	87.39	99.14	241.85
Inputs using sym5 cDs ($j = 4$).	97.47	96.32	96.17	98.75	72.93
Sigmoid MLP(64, 32, 8, 1)					
Inputs without DWT.	50.01	41.24	18.06	82.0	7.86
Inputs using coif5 cAs ($J = 6$).	50.68	39.66	23.51	77.98	0.43
Relu MLP(64, 32, 8, 1)					
Inputs without DWT.	79.92	72.81	59.48	99.96	16.7
Inputs using sym5 cDs ($j = 4$).	97.48	96.66	96.57	98.38	7.16
LS					
Inputs without DWT.	37.99	13.65	72.22	4.34	10.57
Inputs using sym5 cDs ($j = 6$).	49.62	0	99.89	0	0.35
Random Forests					
Inputs without DWT.	97.82	97.25	97.17	98.45	9.42
Inputs using db1 cAs ($J = 4$).	98.08	97.49	97.41	98.73	1.16
CNN					
Inputs without DWT.	97.68	99.94	97.68	95.48	54.17
Inputs using sym5 cDs ($j = 1$).	99.13	99.16	99.13	99.09	22.93
Naive Bayes					
Inputs without DWT.	94.75	90.81	92.42	99.67	8.8
Inputs using bior2.4 cDs ($j = 6$).	95.95	93.37	92.85	98.99	0.35
SVM					
Inputs without DWT.	93.76	88.99	87.41	100	36.31
Inputs using sym5 cDs ($j = 5$).	94.96	90.93	89.84	99.99	1.52

Fig. 6 3-median dataset experimental findings, averaging 100 training and testing processes for each ML model. The ML model inputs are based on the use of Discrete Wavelet Transform (DWT) coefficients with various wavelets and decomposition levels, but not on the application of DWT to the binned light curves [2].

Table 1. Results on simulated data

AUC	Recall	Precision	Accuracy
0.92	0.94	0.92	0.91

Fig. 7 AUC, Recall, Precision and Accuracy for the GBT model [7].

practical application. Comparisons with state-of-the-art deep learning models supported the model's ability, particularly in Kepler data, while underlining its precise advantage in TESS data. However, it is critical to recognize that, while promising, these machine learning-based solutions are not yet sufficiently robust for widespread application. Prudent use with human supervision is still required to reduce potential inaccuracies in unseen data. However, at this point, these models provide a highly reliable system for streamlining the process, considerably lowering the manual review workload for any false positives [7].

The analysis of the Gaussian Process Classifier (GPC) model's performance reveals distinct scoring patterns for various exoplanetary scenarios. The model demonstrates high scores for planets and low scores for eclipsing binaries (EBs) and circumbinary planets (CEBs). Particularly, the model is insensitive to false positives (FPs) resembling brown dwarfs (BDs), ensuring accurate validation. Notably, giant planets tend to receive relatively lower scores, serving as a precaution against the erroneous validation of brown dwarfs. Comparison with the VESPA method showcases substantial classification agreement, especially for confident planet or FP identifications. However, caution is advised regarding intermediate scores and their use in occurrence rate studies. The study underscores the importance of statistical exoplanet validation in the discovery process, especially in the context of faint host stars or challenging candidates. While acknowledging the implicit assumptions and limitations of the methodology, including the reliance on training set coverage and previous dispositions, the study highlights the versatility and value of alternative validation methods. The need for careful consideration of outliers, calibration precision, planet multiplicity, and the specificity of the methodology to the Kepler dataset is emphasized. The intermodel divergence observed suggests the prudence of employing multiple models for validation decisions, thus enhancing the robustness and reliability of the validation process. Overall, this analytical assessment provides insights into the model's effectiveness and limitations, informing future applications and refinements in exoplanetary validation efforts [8].

The Kepler dataset was used to forecast the presence of potential exoplanets using classification approaches including decision trees, random forests, naive Bayes, and neural networks. Furthermore, by clustering those in the same cluster as Earth, the confirmed exoplanets dataset was used to find potentially habitable exoplanets. To improve data visualization and gain a thorough understanding of the samples, the supervised learning task required intensive data cleaning and exploratory data analysis (EDA). The use of feature selection based on correlation resulted in the reduction of redundant features. The chosen machine learning models were then fine-tuned by optimizing hyperparameters experimentally, yielding high accuracies ranging from 88.50% to 99.79%. These models were then compared using 10-fold cross-validation, with the random forest model proving the most effective. Similar EDA and feature selection techniques were used in unsupervised learning, and Earth's characteristics were included in the dataset before clustering. This approach, which divided all exoplanets into 100 clusters, successfully identified exoplanets most likely to be habitable inside the cluster containing Earth. Exploring feature engineering strategies to develop new features as a future extension could potentially improve model performance. Investigating receiver operating characteristics (ROC) and precision-recall curves to assess the diagnostic capabilities of various machine learning models would further add depth to the investigation. Incorporating McNamara's test for algorithm comparison, talking about statistical significance tests, and taking execution time into account in model evaluation would provide full insights [9].

When the results of the Random Forest Classifier (RFC) and Convolutional Neural Networks (CNNs) are compared, it is clear that the correct prediction rate remains steady at around 90% for both methods. Exploring the original light curves for true positives and false negatives, on the other hand, reveals significant trends among the various machine learning algorithms. The RFC tends to discover candidates with typical exoplanetary properties by utilizing attributes obtained from fitted light-curve parameters and external catalog information but not the light curves themselves. However, due to probable noise and gaps in the WASP data, the computed 'best fit' planet characteristics occasionally correlate with the known distribution, even when there is no periodic transiting signal. CNN, on the other hand, approaches transit identification in a fundamentally different way by directly accessing magnitude data folded on the best-fit period. The method in this example focuses on pattern matching to find the right transit form. Examining true positives and false negatives for this subset of data reveals a distinct mechanism for misidentified planets, which is frequently related to misinterpretations of light curves that resemble planets but, when additional information such as transit depth is considered, are likely eclipsing binaries or false positives. Furthermore, when the data is binned down, fainter objects with noisier and irregular signals can resemble transit-like patterns. Systematic disruptions generated by star drift over the CCD can cause transit-like appearances in light curves to be introduced to the CNN. Notably, while the CNN appears promising when used alone, it may overlook certain planets that the RFC can identify and occasionally admit misleading signals, highlighting the importance of combining approaches for optimal results. In practice, the true follow-up rate for these methods would be lower, especially for objects that strongly resemble planets but require additional observation for confirmation. In practice, machine learning findings are generally combined with expert opinion to choose the most scientifically appealing targets for follow-up observations, highlighting the significance of human engagement in target curation. Furthermore, advances in knowledge regarding star parallaxes and radius estimations made possible by missions

such as Gaia assist greatly in improving dwarf/giant discrimination and refining the classification process, particularly in removing some blended eclipse events [10].

V. Problem Statement

The classification of exoplanet candidates remains an important challenge in astronomical research, particularly in distinguishing genuine planetary candidates from false positives. An increase in false-positive classifications can complicate the validation process and affect the reliability of candidate identification. Machine learning provides an opportunity to analyse the available astronomical data and identify patterns that can support this classification process.

This study focuses on the classification of Kepler Objects of Interest (KOIs) using the Kepler cumulative catalogue. The objective is to investigate and compare different machine learning and deep learning approaches for distinguishing planetary candidates from false positives based on the available catalogue attributes. By evaluating the performance of these models using appropriate classification metrics, this study aims to identify approaches that provide reliable classification performance while reducing misclassification. The work also considers the importance of appropriate data preprocessing and model evaluation in assessing the generalisation capabilities of the selected approaches.

VI. IMPLEMENTATION

A. Dataset Description

The experiments were conducted using the Kepler Objects of Interest (KOI) cumulative catalogue obtained from the NASA Exoplanet Archive. The dataset consisted of 9,564 entries and 50 attributes. The target variable, `koi_pdisposition`, was used to classify the objects into two categories: FALSE POSITIVE and CANDIDATE. These classes were encoded as 0 and 1, respectively. The dataset contains several attributes describing the characteristics of the observed objects, including orbital parameters, stellar properties, and transit-related measurements. The `kepid` attribute was retained separately for group-based data splitting but was not used as a predictive feature.

B. Data Preprocessing

The dataset was examined for missing values, irrelevant attributes, and potential sources of data leakage. Identifier and metadata columns, including `rowid`, `kepoi_name`, and `kepler_name`, were excluded from the predictive feature set. Columns containing information that could directly reveal or encode the target disposition, including `koi_disposition`, `koi_score`, and the false-positive flag attributes, were also removed. Only numerical attributes were retained for model development, and columns containing entirely missing values were eliminated. Missing numerical values were handled using median imputation, followed by feature standardisation using StandardScaler.

To prevent data leakage, imputation and scaling were performed within the model pipelines. The preprocessing transformations were fitted using the training data and subsequently applied to the validation and test data.

C. Model Development and Evaluation Strategy

The dataset was divided into training and testing subsets using StratifiedGroupKFold with five splits. The grouping variable was `kepid`, ensuring that observations belonging to the same Kepler target were not distributed across the training and testing subsets. The classical machine learning classifiers were evaluated using five-fold StratifiedGroupKFold cross-validation on the training subset. The evaluated models included Logistic Regression, Decision Tree, Random Forest, Gaussian Naive Bayes, Extra Tree, HistGradientBoosting, AdaBoost, K-Nearest Neighbours, XGBoost, XGBoost Random Forest, LightGBM, and CatBoost. The models were compared using accuracy, precision, recall, F1-score, ROC-AUC, and average precision. CatBoost achieved the highest mean cross-validation F1-score among the evaluated classical classifiers and was therefore selected for final evaluation on the held-out test set.

D. CatBoost Classifier

CatBoost, a gradient boosting algorithm based on decision trees, was selected as the best-performing classical machine learning classifier according to the cross-validation F1-score. The model was configured with 400 iterations, a learning rate of 0.05, and a tree depth of 6. Balanced class weights were used to account for the distribution of the target classes. The model was trained on the complete training subset and evaluated on the held-out test subset.

CatBoost achieved a mean cross-validation F1-score of 0.8398 and a ROC-AUC score of 0.9246. Its final test-set performance is presented in the Results and Analysis section.

E. Multilayer Perceptron

A Multilayer Perceptron (MLP) was implemented using TensorFlow and Keras to evaluate a neural-network-based approach to the classification task. The network consisted of an input layer, followed by two fully connected hidden layers containing 128 and 64 neurons, respectively. ReLU activation functions were used in both hidden layers. Dropout layers with rates of 0.30 and 0.20 were incorporated to reduce overfitting. The output layer consisted of a single neuron with sigmoid activation for binary classification. The model was trained using the Adam optimiser and binary cross-entropy loss. The training process used a batch size of 32 and a maximum of 100 epochs. Early stopping was implemented by monitoring validation loss, with a patience of eight epochs and restoration of the best model weights. Median imputation and feature standardisation were applied before training. The validation subset was created from the training data using group-aware splitting, while the held-out test subset was retained for final evaluation.

	Training Accuracy	Testing Accuracy
xgboost	1.000000	0.990941
Lightgbm	1.000000	0.990244
Hist_Gradient_Boosting	1.000000	0.989547
Random_Forest	1.000000	0.988502
Decision_Tree	1.000000	0.984669
Extra_Tree	1.000000	0.962369
catboost	0.999851	0.989895
XGboostRF	0.997610	0.990941
adaboost	0.996863	0.988850
KNN	0.831192	0.770383
Logistic_Regression	0.750822	0.751916
Gaussian_NB	0.597251	0.593031

	Training Accuracy	Testing Accuracy	CV_Accuracy_Mean	CV_Accuracy_Std
xgboost	1.000000	0.990941	0.990588	0.001303
Lightgbm	1.000000	0.990244	0.991186	0.001595
Hist_Gradient_Boosting	1.000000	0.989547	0.991635	0.001522
Random_Forest	1.000000	0.988502	0.988198	0.001371
Decision_Tree	1.000000	0.984669	0.984912	0.004287
Extra_Tree	1.000000	0.962369	0.974307	0.007200
catboost	0.999851	0.989895	0.992830	0.001117
XGboostRF	0.997610	0.990941	0.989244	0.001743
adaboost	0.996863	0.988850	0.991186	0.002186
KNN	0.831192	0.770383	0.754108	0.009466
Logistic_Regression	0.750822	0.751916	0.757846	0.012857
Gaussian_NB	0.597251	0.593031	0.597104	0.012468

Fig. 8 Training and Testing accuracy before Cross-validation and after Cross-validation of various classification algorithms.

F. Code Availability and Reproducibility

The source code and implementation of this study are publicly available on GitHub. The repository contains the code used for data preprocessing, model development, training, and evaluation.

The complete implementation can be accessed at:

https://github.com/Tanisha6996/Beyond-Boundaries

The repository is provided to support transparency and reproducibility of the experiments presented in this study.

G. Performance Evaluation

The performance of the classifiers was assessed using accuracy, precision, recall, F1-score, ROC-AUC, and precision-recall AUC. Confusion matrices and classification reports were also generated to examine the classification performance across the two target classes.

The classical machine learning models were compared using their mean five-fold cross-validation results. The selected CatBoost classifier and the MLP were subsequently evaluated on the held-out test subset. Since the MLP was evaluated separately rather than through the same cross-validation comparison, its results are treated as a separate neural-network comparison.

VII. Conclusion

The integration of machine learning in the search for exoplanets is revolutionizing astronomy and astrophysics, significantly enhancing the detection and characterization of distant worlds beyond our solar system. This extensive literature review underscores the critical role of machine learning algorithms in interpreting astronomical data, particularly in handling large datasets and identifying subtle signals indicative of exoplanetary existence. Supervised learning algorithms, for instance, excel in classification tasks, effectively distinguishing exoplanetary signals from noise and false positives. Unsupervised learning methods, meanwhile, are invaluable for clustering and pattern identification, revealing underlying structures within datasets essential for exoplanet identification. One of the primary strengths of machine learning is its capability to process vast amounts of data generated by space missions like Kepler and TESS. These missions produce enormous volumes of photometric and spectroscopic data, necessitating automated and efficient analysis methods. Machine learning meets this need by enabling computerized data processing, rapid identification of exoplanetary candidates, and the discovery of elusive, Earth-like exoplanets within habitable zones. Additionally, machine learning enhances the accuracy and precision of parameter estimation for known exoplanets, deepening our understanding of their characteristics. Moreover, the application of artificial intelligence in exoplanet hunting facilitates real-time data analysis, crucial for missions requiring prompt decision-making. The deployment of machine learning techniques in exoplanet research has significantly advanced the field, enabling the study of extensive datasets and leading to the discovery and characterization of numerous exoplanetary systems. As machine learning methodologies continue to evolve and new astronomical missions

are launched, the future promises even more sophisticated applications. These advancements will further unravel cosmic mysteries and potentially lead to the discovery of planets capable of supporting life beyond our solar system.

ACKNOWLEDGMENT

I am grateful to our PG Coordinator and my research guide, Dr. Chandra J, for her essential help and direction throughout this research project. In addition, I would also like to offer my heartfelt gratitude to the scientists and researchers whose scholarly contributions laid the groundwork for my research. Their pioneering efforts considerably improved the theoretical basis of this subject. Each of them has played a pivotal role in the development and realization of this research, for which I am truly grateful.

REFERENCES

[1] C. J. Fluke and C. Jacobs, "Surveying the reach and maturity of machine learning and artificial intelligence in astronomy," *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, vol. 10, no. 2, p. e1349, 2020.

[2] M. Jara-Maldonado, V. Alarcon-Aquino, R. Rosas-Romero, O. Starostenko, and J. M. Ramirez-Cortes, "Transiting exoplanet discovery using machine learning techniques: A survey," *Earth Science Informatics*, vol. 13, pp. 573–600, 2020.

[3] K. A. Pearson, L. Palafox, and C. A. Griffith, "Searching for exoplanets using artificial intelligence," *Monthly Notices of the Royal Astronomical Society*, vol. 474, no. 1, pp. 478–491, 2018.

[4] D. Baron, "Machine learning in astronomy: A practical overview," *arXiv preprint arXiv:1904.07248*, 2019.

[5] S. E. Thompson, F. Mullally, J. Coughlin, J. L. Christiansen, C. E. Henze, M. R. Haas, and C. J. Burke, "A machine learning technique to identify transit shaped signals," *The Astrophysical Journal*, vol. 812, no. 1, p. 46, 2015.

[6] P. Chintarungruangchai and G. Jiang, "Detecting exoplanet transits through machine-learning techniques with convolutional neural networks," *Publications of the Astronomical Society of the Pacific*, vol. 131, no. 1000, p. 064502, 2019.

[7] A. Malik, B. P. Moster, and C. Obermeier, "Exoplanet detection using machine learning," *Monthly Notices of the Royal Astronomical Society*, vol. 513, no. 4, pp. 5505–5516, 2022.

[8] D. J. Armstrong, J. Gamper, and T. Damoulas, "Exoplanet validation with machine learning: 50 new validated kepler planets," *Monthly Notices of the Royal Astronomical Society*, vol. 504, no. 4, pp. 5327–5344, 2021.

[9] Y. Jin, L. Yang, and C.-E. Chiang, "Identifying exoplanets with machine learning methods: A preliminary study," *International Journal on Cybernetics &amp Informatics*, vol. 11, pp. 31–42, apr 2022.

[10] N. Schanche, A. C. Cameron, G. Hébrard, L. Nielsen, A. Triaud, J. Almenara, K. Alsubai, D. Anderson, D. Armstrong, S. Barros, *et al.*, "Machine-learning approaches to exoplanet transit detection and candidate validation in wide-field ground-based surveys," *Monthly Notices of the Royal Astronomical Society*, vol. 483, no. 4, pp. 5534–5547, 2019.

[11] M. Ansdell, Y. Ioannou, H. P. Osborn, M. Sasdelli, J. C. Smith, D. Caldwell, J. M. Jenkins, C. Räissi, D. Angerhausen, *et al.*, "Scientific domain knowledge improves exoplanet transit classification with deep learning," *The Astrophysical journal letters*, vol. 869, no. 1, p. L7, 2018.

[12] K. H. Yip, Q. Changeat, N. Nikolaou, M. Morvan, B. Edwards, I. P. Waldmann, and G. Tinetti, "Peeking inside the black box: Interpreting deep-learning models for exoplanet atmospheric retrievals," *The Astronomical Journal*, vol. 162, no. 5, p. 195, 2021.

[13] S. Cuéllar, P. Granados, E. Fabregas, M. Curé, H. Vargas, S. Dormido-Canto, and G. Farias, "Deep learning exoplanets detection by combining real and synthetic data," *Plos one*, vol. 17, no. 5, p. e0268199, 2022.

[14] S. Zucker and R. Giryes, "Shallow transits—deep learning. i. feasibility study of deep learning to detect periodic transits of exoplanets," *The Astronomical Journal*, vol. 155, no. 4, p. 147, 2018.

[15] L. Yu, A. Vanderburg, C. Huang, C. J. Shallue, I. J. Crossfield, B. S. Gaudi, T. Daylan, A. Dattilo, D. J. Armstrong, G. R. Ricker, *et al.*, "Identifying exoplanets with deep learning. iii. automated triage and vetting of tess candidates," *The Astronomical Journal*, vol. 158, no. 1, p. 25, 2019.

[16] H. Valizadegan, M. J. Martinho, L. S. Wilkens, J. M. Jenkins, J. C. Smith, D. A. Caldwell, J. D. Twicken, P. C. Gerum, N. Walia, K. Hausknecht, *et al.*, "Exominer: A highly accurate and explainable deep learning classifier that validates 301 new exoplanets," *The Astrophysical Journal*, vol. 926, no. 2, p. 120, 2022.